

BINDING ENERGY AND THE WEAK FORCE FROM HQIV COMPOSITE-TRACE WEIGHTS: LOCK-IN MASSES, THREE LEDGERS, AND ZERO-KNOB HALF-LIVES

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ABSTRACT. **Position in the publication chain.** Companion to the published TUFT+SM synthesis [10] (DOI [10.5281/zenodo.20601215](https://doi.org/10.5281/zenodo.20601215)). This is the seventh paper in the HQIV preprint series and the third Tier-2 Standard-Model layer entry *after* the published TUFT+SM synthesis [10]. Tier 0–1 fix the discrete spine [12, 14, 16, 18, 15, 13, 17]. **This note establishes** the local-curvature slot equations—named $B_{\text{curv}}(\xi)$, three β ledgers, slot-wise mass/binding closure, and outside-contact primitives—witnessed from nucleon binding through bulk condensed matter. **Companion** Tier 2 notes on baryogenesis lock-in [9] (published: DOI [10.5281/zenodo.20711255](https://doi.org/10.5281/zenodo.20711255)) and BBN abundances [8] apply the *same* equations ($\xi(T)$, outside release, opacity slots, dynamic- C_2 lapse clock) at cosmological epoch; they do not introduce separate physics layers (§11). Outside temperature and gravity cite [17, 7].

Result (masses). At lock-in shell reference $M = 4$ proton and neutron masses are constituent energies minus a shared 8×8 composite-trace binding E_{bind} on the $\mathfrak{so}(8)$ network. Isospin splitting is constituent bookkeeping; machine-checked identities include positivity, shared-surface splitting, and meta-harmonic alignment (derived nucleon mass pack).

Result (β tipping, v4). Three ledgers remain strictly separate: (i) strong/curvature overlap residual, (ii) kinematic endpoint Q from the dynamic mass budget minus m_e , (iii) weak width from Fermi coupling from beta with a generic valley/caustic geometry factor for bonded clusters and a *local-curvature neutrino-opacity catalysis* on the free branch (§9.4). The free-neutron readout matches reference weak *width* to $\lesssim 0.01\%$ in effective phase space ($\Delta Q_{\text{phase}} \approx 0.02$ keV on $Q_{\text{phase}} \approx 782$ keV; $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$) with **no** half-life fit parameters. Integrated clock time $\tau_n \approx 880$ s at 300 K ($\approx 1.00 \times$ the 879 s reference band; envelope 875–884 s brackets bottle and beam representatives) uses the same $B_{\text{curv}}(\xi)$ stack as bound-state and BBN integrator witnesses. Light-isotope masses agree at mean 0.003% on the $A \leq 4$ panel (curvature G_{eff} spine @ ξ_{lock}). The same spine extends through $A = 16$ on the post- α and multi- α ledger with mean $|\Delta B|/B_{\text{PDG}} \approx 1.2\%$ on the full twelve-isotope witness panel ($\approx 1.2\%$ for $A > 4$ alone; Table 3). Tritium β^- : the width ledger requires a distinct interior valence well (mass and width slots cannot collapse—radical ledger locality); with slot-wise local curvature closure on each mass and binding budget, $\tau_T/\tau_{\text{ref}} = 1.000$ on the same equations that fix M_T (relic-bath catalysis is *shielded* for $A > 1$). Post- α EC/ β^+ on ${}^7\text{Be}$ books facet-proton *local contact* mass for endpoint Q and the same post- α outside-valley count for the weak width ledger (decay-chain calculator). ${}^8\text{Be}$ α breakup discharges through the two- α trapped-inside saddle and width ledger ($\tau \approx 5.5 \times 10^{-17}$ s witness vs. reference $\sim 6.7 \times 10^{-17}$ s). Together these validate the local-curvature spine against laboratory readouts.

Outside environment. Lab readouts book temperature through $T(\xi) = T_{\text{P1}}/\xi$ and a cumulative weak-field binding stack (Earth + Sun @ 1 AU + Galactic v_c^2/c^2) folded into outside G-eff support. Galactic binding dominates the gravity ledger ($\sim 98\%$ of ε_g); Earth-surface labs are blind to it at sub-ppm on τ_n , while BBN at $T \sim 0.1$ MeV sees outside weakening at the percent level—feeding the coupled neutron inventory and ${}^7\text{Li}/H$ witnesses documented in [8].

Single absolute anchor (strictly isolated). Laboratory proton mass 938.272 MeV appears only as a lapse-readout anchor for comparison, not as an input to E_{bind} or weak widths. Reference half-lives and isotope masses in tables are comparison targets only.

Status. Proved: composite-trace binding spine, endpoint- Q identities, weak-width slot wiring (upstream [10]), Fano/Hopf bridge shape, outside-gravity modulator naming, local-curvature neutrino-opacity definitions (§9.4). Strong binding and confinement language discharge through the same Lagrangian + [11], not legacy QCD inputs. Computed witnesses (Python): light-panel masses, τ_n , tritium width, temperature/gravity sweeps, BBN integrator opacity export (§2.1, §10); **derived allotropes** and bulk geometry from molecular inputs via the `hqiv_lab` package (§8.3, §8.2); condensed-phase ρ_{curv} , melt, and material response on the same spine (raw contact lattice vs. medium-closed bulk readout); response slots use standard continuum formulae [27, 2, 3]. Not re-derived here: full β spectra; dynamical SU(3); actinide fission decay chains; MSbar current-quark

inputs; sub-leading non-contact network refinements (conjecture in §2). **Extended binding panel** ($A \leq 16$, Table 3) is a computed witness on the same spine; **post- α β^+ /EC and two- α width** witnesses are wired on the decay-chain calculator (§7.8); full actinide α ladders remain open. **Light-element BBN** (D, ${}^3\text{He}$, ${}^7\text{Li}$) is a computed witness via the faithful integrator (§2.1, Appendix B); ${}^7\text{Li}$ uses the ${}^3\text{He}+{}^4\text{He}\rightarrow{}^7\text{Be}\rightarrow{}^7\text{Li}$ network ladder with trimer-width, stoichiometric D-budget, and synthesis-window tail-gate slots (BBN stoichiometric integrator).

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as Fin 4, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation ($Q_L, u^c, d^c, L, e^c, \nu^c$), the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite Fin 3 label sum of the same cancelled one-generation block, not a fitted multiplicity; see Appendix A, anomaly cancellation.
- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish; see patch topology discharge.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). ****Matrix Wilson layer:**** full four-edge $SO(8)$ generator transport and Hopf-shell T^4 torsion plaquettes with pinned non-abelian witness ($SO(8)$ plaquette holonomy, witness `so8PlaquetteHolonomyDischarged_holds`). ****Wilson-kinetic equivalence**** (Wilson-kinetic glue, witness `wilsonKineticPlaquetteEquivalence_discharged`). ****Partial / open:**** Haar measure on rotated charts (non-abelian Haar measure scaffold); general flat HQVM kinetic boost and full strong-interacting discrete-action Poincaré (discrete action Poincaré programme). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart; optional strong-colour certificate in the Lean catalog.
- **Rapidity phase and Lorentz invariance.** The null-lattice carrier `NullLatticeEvent` charts into 1+1 Minkowski space by the classical rapidity parametrization $\Lambda(\eta)(1, 1)$; rapidity addition on the carrier is equivariant under `boostMatrix11`, the Minkowski pairing is bilinearly invariant, and the polar-angle / `zetaHQIVTerm` phase is unchanged when ϕ transforms as a Lorentz scalar. The honest 3+1 extension preserves `minkowskiSq4` on the embedded $(t, x^1, 0, 0)$ plane.
- **Spatial rotations and directional readouts.** Orthogonal $O(3)$ rotations on Fin 3 spatial vectors preserve Euclidean inner products, $\|\mathbf{r} \times \mathbf{v}\|^2$, axis-projection fractions, and the spatial block of `minkowskiSq4` when time is fixed (spatial-rotation Lorentz closure, witness `spatial_rotation_lorentz_closure_discharged`). Applied discharge: CMB sky rotation shifts

the Stokes angle additively while preserving $\cos^2 + \sin^2 = 1$; O-Maxwell fluid sources rotate covariantly with gradients; coronal axis projections are joint rotation invariants. Full patch Lorentz on the flat chart is the rapidity bundle plus this spatial bundle (witness `full_lorentz_closure_discharged`). Continuum interacting QFT / Wightman theorems are not claimed.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

Projection geometry and the classical ζ dictionary (2026-06). HQIV uses *zeta* in two distinct, now formally separated roles:

- **Sector zeta on the patch carrier** (TUFT+SM synthesis): fibre-torsion subleading bodies, `zetaHQIVTerm` phase on Hopf shells, and holonomy discharge bundles on the octonion light-cone carrier—finite spectral readouts indexed by shell and Fano sector, not the Riemann function.
- **Classical $\zeta(s)$ as an $\text{SO}(4)$ projection readout** ([19]): the Mathlib Riemann zeta packaged as a *regional* closed form on $\mathbb{C}$ —Dirichlet/Euler for $\Re s > 1$, $\sin\text{--}\cos\text{--}\Gamma$ assembly on the strip, Bernoulli slots—with the functional-equation pair $(\sigma, 1 - \sigma)$ interpreted as a 45° quaternionic projection. Machine-checked on this spine: harmonic divergence forces Δ and $\mathfrak{so}(4)$ closure; shell sum $= \zeta(s)$ for $\Re s > 1$; polar decoupling $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (modulus σ -only, phase t -only); prime-log Diophantine independence; Goldbach circle $pq + (q - N)^2 = N^2$ as additive curvature in the log channel; and a large family of RH-*equivalent* locators (twiddle, spectral weights, polar collapse, transformer adjoint law, S^7 torsion cancellation) with *zero slack*—not a discharge of RH or Goldbach parity.

Continuum heat-flow and de Bruijn–Newman notation in the same story spine are *comparison ports* (`dbnHeatFamily`, lattice dominance hypotheses); they are not load-bearing continuum extensions of the patch. Formal mathematical notes may include only the readout-dictionary paragraph (`readout_dictionary_messaging.tex`) rather than this full contract when the manuscript is projection geometry rather than patch phenomenology.

1. INTRODUCTION

Nuclear binding energy is a small fraction of the nucleon mass budget—typically ~ 8 MeV per nucleon, or $\lesssim 1\%$ of the ~ 940 MeV rest scale—yet it governs almost every macroscopic consequence of strong interaction physics. Whether a nucleus exists at all, how light elements freeze out in big-bang nucleosynthesis (BBN), how stars burn through the main sequence, and why the free neutron decays in ~ 880 s while the proton appears stable are all questions in which *binding* and *weak tipping* enter on equal footing with mass accounting. In conventional treatments the binding problem is delegated to effective nuclear potentials, chiral EFT, or—at the fundamental level—lattice QCD numerics that are accurate but opaque as a *geometric* story. The weak lifetime problem is usually separated further: conventional treatments import Fermi theory or full electroweak matrix elements for partial widths, while the neutron bottle–beam tension remains an experimental methodology puzzle layered on top. HQIV does not add a third ad hoc weak model here: the electroweak width slot is already packaged in the TUFT+SM Lagrangian synthesis [10]; this note applies that bridge to nucleon binding and the light-isotope β panel (Remark 7.4).

The Horizon-Quantized Informational Vacuum (HQIV) programme asks whether the same discrete spine that fixes gauge structure and fermion masses can also *name* binding and β -decay through **local curvature slots**—each contact, ledger, and isotope budget carries its own causal closure on the null lattice, with no global potential fits. The present note is the Tier-2 foundation that proves and witnesses that spine from lock-in nucleons through light isotopes and laboratory condensed phases (water as the primary melt witness; §8.2), with explicit Lean certificates and reproducible Python readouts. Baryogenesis and BBN companions inherit the identical slot equations, evaluated at cosmological $\xi(T)$ rather than Earth-lab lock-in.

Minimal axioms and where to enter the series. HQIV rests on three intertwined commitments, all machine-checked in the public Lean library [20]:

- (1) a **discrete null lattice** with octonionic eightfold mode counting and cumulative simplex bookkeeping;
- (2) **informational monogamy** (horizon-limited causal capacity) forcing the imprint pair $(\alpha, \gamma) = (3/5, 2/5)$ at every truncation;
- (3) an **octonionic $\mathbb{R}^8$ carrier** whose $\mathfrak{so}(8)$ network weights supply sector projections including electroweak kinetic terms.

New readers should start with the framework overview on Zenodo [12] (concept DOI 10.5281/zenodo.18794889; latest version [12]), then the Tier-0 light-cone and $\text{SO}(8)$ closure records [14, 16], the Tier-1 variational spine [18, 15, 13, 17], and the published Tier-2 TUFT+SM synthesis [10]. The baryogenesis and BBN companion notes apply the same local-curvature equations at epoch-dependent $\xi(T)$; they do not introduce alternate binding rules. Scope and contributions. We do **not** reproduce continuum QCD or a full shell-model chart. We **do** prove radical locality on the discrete action: every load-bearing quantity is a *named slot* closed by local curvature on its contact network, with laboratory data used only to **validate** the slot readouts, never as fit inputs.

- **Local curvature from nucleons to condensed matter.** The same outside-contact and $B_{\text{curv}}(\xi)$ primitives run from composite-trace binding (§3–§5), through three strict β ledgers and valley geometry (§7, §8), to molecular hosts and bulk ρ_{curv} (§8.2, §8.2)—one causal spine, three scales, zero alternate equations.
- **Geometric binding.** Proton and neutron masses are constituent energies minus shared E_{bind} on the $\mathfrak{so}(8)$ network; cluster binding follows the curvature + G_{eff} shared-well network spine (inside trapped curvature plus outside ladder, γ -network, barbell, tetrahedral closure) with slot-wise isotope weighting (§4).
- **Lapse renormalization.** Laboratory comparison uses a one-point proton anchor and shell lapse readout (§6). The anchor is strictly isolated: it does not feed E_{bind} or weak widths.
- **β -decay programme.** Weak tipping uses three non-collapsible ledgers plus skew alignment on the octonion carrier (§7). On the $A \leq 4$ panel, witnesses validate free-neutron width to $\lesssim 0.01\%$ in effective phase space, masses at 0.003% mean, and tritium $\tau_T/\tau_{\text{ref}} = 1.000$ from the same slot closure as M_T (Table 7).
- **Lean audit.** Binding positivity, ledger separation, endpoint- Q identities, width-slot wiring, and bottle/beam method structure are sorry-free (Appendix A).
- **Downstream epoch dynamics.** §11 states how baryogenesis and BBN companions reuse these slots with $\xi(T)$ —dynamic in epoch, identical in equation form.

Roadmap. Section 2 states claim discipline and collects headline witness numbers. Sections 3–§6 derive lock-in nucleon masses (mode counts, isospin split, lock-in sensitivity). Section 7 is the β -decay core: skew alignment, ledgers, endpoints, weak width, explicit width assembly (§7.7), Fano/Hopf bridge, and milestone roadmap. §4 constructs the cluster network; §5–§6 fix lock-in masses; §8–§9 treat bonded-cluster geometry, outside temperature and gravity, and local neutrino-opacity catalysis. Section 10 tabulates the light-isotope panel. §11 shows the same local-curvature equations carried

forward to the baryogenesis and BBN companions. Appendices A and B map reader-friendly names to Lean modules and reproducer scripts.

2. SCOPE AND CLAIM DISCIPLINE

The combined TUFT+SM synthesis [10] packages sector projections including the weak kinetic aggregate and establishes the electroweak ladder and Yukawa readouts. **This note answers two questions:**

- (1) How are proton and neutron masses defined from the same 8×8 network binding at lock-in?
- (2) How are β -decay Q -values and half-lives discharged from named weak slots and local curvature on each contact—without importing PDG lifetimes as inputs?

Thesis: local curvature slots and radical locality. HQIV physics is organized as **named slots** on a discrete null lattice: each contact, ledger, and budget carries local curvature closure, not a fitted global potential. The same slot algebra propagates causally from a single nucleon trace ((2)) through cluster Casimir caustic geometry, three non-collapsible β ledgers, molecular outside contacts, and bulk ρ_{curv} (§8.2). Laboratory tables validate the slot readouts; they do not supply tunable parameters. Baryogenesis and BBN companions (§11) change only the epoch witness $\xi(T)$ and outside-release modulator—the equations are unchanged. That continuity is the central claim of this note: *radical locality* from nucleon binding through water, with cosmology as the same spine evaluated hotter and sparser.

Domain: network simplicity through $A = 4$. The witnesses in this note cover the light-isotope ladder through helium-4 (p , n , D , T , ${}^3\text{He}$, ${}^4\text{He}$): shared-surface composite-trace binding, valley-count geometry, and interior width-well booking suffice *without* alternate binding rules per nucleus. On that panel the derived network readouts agree with laboratory readouts at mean 0.003%, free-neutron τ_n at $1.00 \times$ reference width, and tritium $\tau_T/\tau_{\text{ref}} = 1.00$ under the same slot-wise local curvature closure as the mass table (Table 7).

Open conjecture (untreated here). For nuclei beyond the $A = 4$ light panel, we *conjecture* that inverse-square weights on the **50**(8) contact network and Casimir caustics on non-contacting nucleon pairs propagate additional structure not required to close the witnesses tabulated here. That programme is **not** executed in this note; we invite independent study without altering the three-ledger spine proved in Lean (§7, Appendix A).

Upstream Lagrangian discharge (no legacy imports). Readers sometimes ask whether this note “still owes” continuum QCD (confinement, asymptotic freedom, a full strong potential), Fermi theory, or fitted condensed-matter potentials. It does not: those are **not** missing ad hoc layers to be patched later. The published TUFT+SM Lagrangian synthesis [10] already projects the discrete octonion action onto $\mathcal{L}_{\text{SM}}$ —strong kinetic aggregate on the four strong channels, electroweak width slot, G_F from β , and leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ wiring. Strong binding on the same 8×8 network is the inner–outer Casimir trapping on those channels, made explicit in the gluon-curvature companion [11] (continuum language of “gluons” and “confinement” is a translation of that geometric trapping, not an independent import). **This note consumes those slots**; it does not re-derive the full Lagrangian bridge, a dynamical non-abelian SU(3) theory, or spectral β shapes here. The upstream synthesis also discharges *finite-patch* gauge consistency: explicit SM anomaly-trace cancellation and single-sector instanton/ θ /Chern/winding obligations (anomaly cancellation, patch topology discharge), plus rapidity-phase and spatial-rotation Lorentz closure on the flat patch chart (rapidity–Lorentz closure, spatial-rotation Lorentz closure), full four-edge SO(8) Wilson plaquettes, abelian Wilson–kinetic equivalence, and a partial strong-interacting discrete-action Poincaré bundle (SO(8) plaquette holonomy, action holonomy glue, discrete action Poincaré programme)—not a continuum path-integral or smooth-bundle upgrade. What remains open on the discrete spine are *extensions* of the same upstream bridge—Haar measure on rotated charts, general flat HQVM kinetic invariance

under the strong-Poincaré bridge, forbidden transitions, radiative normalization, higher- A non-contact propagation (§2), and a derived lock-in principle for m_{lockin} without the proton anchor—not replacement of the Lagrangian by legacy effective potentials or lattice tables.

Numerical domain (where witnesses are sharp vs. structural). On the $A \leq 4$ nuclear panel, neutron width and light-isotope masses are the primary quantitative successes (Remark 2.1, Table 7). Condensed-phase readouts in §8.2 are **structural witnesses** on the same outside-contact spine: a raw contact-lattice step can sit several percent below tabulated bulk density until ambient liquid-water curvature (surrounding homogeneous medium) is booked; the closed pipeline then lands melt, refractive index, and bulk thermodynamics in good agreement with laboratory water/ice (§8.2, Table 4). Meta-harmonic nucleon readouts several percent above PDG remain an acknowledged dressing + derived-lock-in problem on the upstream mass ladder, not a contradiction of Theorem 3.2.

Claim status is summarised in Table 1; headline numbers are in Remark 2.1.

TABLE 1. Claim map: proved in Lean vs computed witnesses vs not claimed.

Topic	Lean status	Witness
Composite-trace E_{bind}	proved	§3
Strong binding / inner Casimir trapping	upstream Lagrangian + [11]	§2
Wilson–kinetic + partial strong Poincaré	partial (upstream)	§2
Cluster network + per-isotope BE weighting	structural + witness	§4
Derived $M_p, M_n, \Delta M$	proved	§5
Three β ledgers	definitions + identities	§7
Endpoint $Q_{\beta^-} = \Delta M - m_e$	proved	§7
Weak width slot (G_F , matrix element)	proved wiring (upstream)	§7
Explicit Γ assembly (referee stack)	defined + witnessed	§7.7
β spectral shapes / forbidden / radiative	upstream extension milestone	§7.8
Local ν -opacity catalysis (B_{curv})	defined + OOM theorem	§9.4
Valley geometry factor	defined	§8
Fano/Hopf bridge energy	proved shape	§7.6
Outside $T(\xi)$ + gravity ε_g	partial Lean + Python	§9
Bottle/beam ledger (weak Ledger III)	proved structure	§10
β width qualification	proved wiring	§7
Allotrope from bonds + VSEPR motif	structural Lean + <code>hqiv.lab</code>	§8.3
ρ_{curv} , melt, n , k_{th} , C_p	witness (raw + medium-closed)	§8.2
$\tau_n \approx 880$ s, light masses	witness ($A \leq 4$ panel)	§10
Higher-order non-contact network / Casimir caustics	conjecture (untreated)	§2
Full nuclear chart beyond light isotopes	not claimed	—
${}^7\text{Li}$ / BBN light-element integrator	structural Lean + witness	§2.1

Remark 2.1 (Computational witnesses (lock-in panel)). *All entries are computed witnesses on the (α, γ) ladder at lock-in; reference masses and half-lives are comparison targets only (no fit knobs). Recompute with the scripts in Appendix B.*

- **Free neutron** @ 300 K, full gravity stack: $\tau_n \approx 880$ s ($\approx 1.00 \times$ the 879 s band); width ledger $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$, $\Delta Q_{\text{phase}}/Q_{\text{phase}} \lesssim 0.01\%$.
- **Light-isotope masses**: mean $|\Delta M|/M_{\text{ref}} \approx 0.003\%$ on the $p, n, D, T, {}^3\text{He}, {}^4\text{He}$ panel (curvature spine; Table 7).
- **Tritium** β^- : ledger separation (mass vs width wells) is mandatory for causal locality; slot-wise local curvature on each budget yields $\tau_T/\tau_{\text{ref}} = 1.000$ on the same equations as M_T (§8, Table 7).
- **BBN light elements** @ η_{paper} (faithful integrator witness; [8]): $Y_p \approx 0.244$ (post-synthesis neutron inventory; obs. 0.244 ± 0.004); $D/H \approx 2.51 \times 10^{-5}$ (synthesis window + $n+p \rightarrow D$ + coupled inventory + $\text{bbnSynthesisWindowTailGate}$; obs. $(2.53 \pm 0.04) \times 10^{-5}$, $z \approx -0.5$); ${}^3\text{He}/H \approx 1.01 \times 10^{-5}$; ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-depletion; Coc semi-analytic comparison 4.65×10^{-10}) (Spite-plateau band $(1.6\text{--}4.5) \times 10^{-10}$ via ${}^3\text{He}$ stratification gates). Recompute: [BBN integrator script](#); tables: [BBN paper tables script](#).
- **Outside environment**: Earth-lab τ_n shift ~ 0.15 ppm from gravity; BBN-relevant $T \sim 0.1$ MeV sees $\sim 22\%$ outside weakening on the free branch.
- **Condensed phase**: `hqiv_lab` derives allotropes and material response on four GMTKN55 motifs (§8.2): mean $|\Delta\rho|/\rho_{\text{NIST}} \approx 0.4\%$, $|\Delta n|/n_{\text{NIST}} \approx 2\%$, $|\Delta T_{\text{sl}}|/T_{\text{NIST}} \lesssim 1\%$ on the $\kappa_6(\xi, \rho_\kappa(n))$ spine—parameter-free vs. periodic PBE-D3 [26, 43, 5] at comparable accuracy on density without per-species crystallographic inputs.

DATA AND CODE AVAILABILITY

Machine-checked proofs build under the `paper_nucleon_binding` Lake target in the public HQIV Lean repository [20]. Quantitative witnesses are reproduced by the Python scripts listed in Appendix B; a self-contained bundle ships on the Zenodo record as `scripts.zip` (see Appendix B and the bundled `README.md` / `MANIFEST.sha256`). The bundle includes the `hqiv_lab` package (`pip install -e .`), isotope/phase/BBN entry scripts, and frozen witness JSON exports. The proton lock-in anchor 938.272 MeV calibrates comparison tables only; it does not enter E_{bind} , width slots, or $B_{\text{curv}}(\xi)$.

3. THE 8×8 COMPOSITE-TRACE BINDING LAW

3.1. Mode counts and the forced imprint $\alpha = 3/5$. Fix a non-negative shell index m . The positive octant slice at shell m ,

$$\{(x, y, z) \in \mathbb{Z}_{\geq 0}^3 : x + y + z = m\},$$

contains exactly the stars-and-bars count $\binom{m+2}{2}$ of non-negative integer solutions. Attaching the octonionic eightfold multiplicity yields the available horizon modes

$$(1) \quad A(m) = 8 \binom{m+2}{2} = 4(m+1)(m+2).$$

Write $N_{\text{simp}}(m) := (m+1)(m+2)$ and the cumulative simplex sum $C(n) := \sum_{m=0}^n N_{\text{simp}}(m)$. Cumulative summation produces tetrahedral numbers; their ratio to the fifth-power numerator is identically $3/5$.

Lemma 3.1 (Cubic cumulant). *For every $n \in \mathbb{N}$, $3C(n) = (n+1)(n+2)(n+3)$.*

Proof. Induction on n . For $n = 0$, $C(0) = N_{\text{simp}}(0) = 2$ and $(0+1)(0+2)(0+3) = 6 = 3C(0)$. If $3C(n) = (n+1)(n+2)(n+3)$, then $C(n+1) = C(n) + N_{\text{simp}}(n+1)$ with $N_{\text{simp}}(n+1) = (n+2)(n+3)$, hence $3C(n+1) = (n+2)(n+3)(n+4)$. Machine-checked as `latticeAlphaRatio.eq_alpha` in [13, 14]. $\square$

Theorem 3.2 (Forced imprint exponent). *With $r(n) := (n+1)(n+2)(n+3)/(5C(n))$, one has $r(n) = 3/5$ for all n . Define $\alpha := 3/5$ and $\gamma := 1 - \alpha = 2/5$.*

Proof. By Lemma 3.1, $(n+1)(n+2)(n+3) = 3C(n)$, hence $r(n) = 3/5$. $\square$

Remark 3.3 (Three naturals + one real). *Under (1), transverse null-mode counting uses three independent $\mathbb{N}_0$ coordinates with $x + y + z = m$ while causal height along maximal chains supplies one ordered $\mathbb{R}$ axis; the Fano/octonion carrier ties internal phase to $\mathbb{F}_2^3$. The full synthesis theorem is in [18]; this note uses only (1) and Theorem 3.2.*

The composite-trace binding law below evaluates at a chosen shell m through the lattice simplex count and $\alpha_{\text{eff}}(m)$; no additional exponent is tuned.

3.2. Network binding functional. The bound-states module defines network binding from generator diagonals $\text{diag}_k(i)$ and an eight-component state ψ_i :

$$(2) \quad w_k = \sum_{i=0}^7 \text{diag}_k(i) \psi_i^2, \quad E_{\text{bind}}(m) = \sum_k w_k \cdot (\text{latticeSimplexCount}(m) \cdot \alpha_{\text{eff}}(m)).$$

The nucleon trace witness uses a triadic state on three carrier slots; triadic triple coupling shows this equals three times the shell coupling at m —the normalisation consumed by BBN and excited-state readouts [8, 10].

Binding is a discrete network sum on the null lattice at shell m ; it is not a continuum QCD potential fit. Strong-channel weights are the projection of the same network onto the octonion strong directions; confinement and asymptotic freedom in continuum language translate to that inner-Casimir trapping on the carrier [11, 10]—discharged upstream, not imported here as a separate legacy model.

4. CONSTRUCTING THE BINDING NETWORK AND PER-ISOTOPE WEIGHTING

4.1. Scale witness and local background curvature. HQIV allows **any** high-precision local quantity to serve as the active *scale witness* that fixes the MeV chart for a given pipeline (scale witness discipline; [10]): default proton lock-in at $\text{referenceM} = 4$, with CODATA- α and CMB-now as explicitly labelled comparison charts. Exactly one witness anchors a solve; all PDG/CODATA rows in the tables are **comparison targets**, not simultaneous fit inputs.

On Earth, tabulated nucleon and light-isotope masses are treated as the *local-curvature closure* of the homogeneous background the bare lock-in derivation does not expand: each isotope slot carries $\kappa_{\text{mass}} = M_{\text{ref}}/M_{\text{derived}}$ as a **local curvature budget**, not a fitted correction (§8, curvature-MassImprint). Laboratory readouts validate the slot algebra; they do not supply simultaneous fit inputs.

Other environments (BBN $T \sim 0.1$ MeV, heat-death/CMB limits, neutron-star lapse $N < 1$ slices) use the *same* slot equations with different outside-curvature modulators $\xi(T)$ (§9, §11).

4.2. From composite trace to cluster binding. Cluster binding is assembled in three layers, all on the $\mathfrak{so}(8)$ network (nuclear curvature binding; HQIV nuclei module):

- (1) **Shared nucleon trace** — $E_{\text{bind}}(\text{referenceM})$ from (2) subtracted equally from p and n (§5).
- (2) **Inside trapped curvature** — $B_{\text{in}} = A \times E_{\text{trace}} \times \Delta(\text{metaHorizonTrappedInsideRatio})$ when the cluster closes onto a readout shell (inside binding).
- (3) **Outside G_{eff} network** — contact units $\times G_{\text{eff}}(\theta/\theta_0) \times E_{\text{trace}}$ on the shared-well spine: intrinsic well deepening D_{intr} from valley occupancy, constructive ladder, γ -network (for $A \geq 3$), barbell torus (for $A \geq 3$), tetrahedral closure (for $A \geq 4$), and post- α ledger for $A > 4$ (outside binding; caustic hierarchy names the geometric sublayers).

Each nucleon books a meta-horizon shell at radius $R_m = m + 1$ (vacuum-mode density witness): available modes $\propto (m+1)(m+2)$ sets the curvature proxy. Pairwise contact uses the valley potential scalar; each constructive rung adds two valley contacts (valley count, proved additive). Casimir caustic layers (sphere-pair, barbell, deepen, tetra) are *diagnostic* outside witnesses (nuclear caustic binding script); PDG-comparable totals $B_{\text{net}} = B_{\text{in}} + B_{\text{out}}$ export from the curvature program (curvature binding program script, [data/curvature.binding.program.json](#)).

4.3. Slot-wise local curvature on each isotope budget. Mass and width ledgers are *distinct local slots*; binding enters each with its own curvature budget:

- **Mass ledger:** symmetric per-nucleon booking B_{cluster}/A enters dynamic p/n readouts; proved ΔM is unchanged (dynamic nucleon PN module).
- **Local curvature on each slot:** each (A, Z) mass budget carries κ_{mass} ; the cluster caustic in the width well carries $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$ —the same slot closure that validates mass tables validates tritium $\tau_T/\tau_{\text{ref}} = 1.000$ (§8).
- **Width ledger:** interior valence well uses $B_{\text{cluster}}/(A-1)$ blended with B_{cluster}/A ; valley exponent $2(A-1)+1$ suppresses weak tipping for bonded clusters.

Witness export: curvature binding program script, nuclear caustic binding script (diagnostic layers), inside/outside binding script, [isotope benchmark JSON](#).

TABLE 2. Light-panel binding @ lock-in (ξ lock-in, $m = 4$): curvature + G_{eff} shared-well network vs PDG total binding (comparison only). Mean $|\Delta B|/B_{\text{PDG}}$ on the $A \leq 4$ rows below compares the curvature G_{eff} binding ledger to PDG total- B tables (reference nucleon masses; $\sim 1.1\%$ on these four isotopes). The authoritative light-panel witness is cluster mass: $|\Delta M|/M_{\text{ref}} \approx 0.003\%$ (Table 7). Extended post- α isotopes ($A > 4$) use the same spine (Table 3; mean $\sim 1.2\%$ on the full panel). Witness: curvature binding program script @ $m = 4$; [data/curvature.binding.program.json](#), [data/binding.energy.program.json](#).

Isotope	A	Valley contacts	B_{in}	B_{out}	B_{net}	B_{PDG}	Δ (%)	Network narrative
D	2	2	0.59	1.60	2.19	2.22	−1.8	Ladder + D_{intr} ; stable.
T	3	4	0.88	7.54	8.42	8.48	−0.7	γ -net + barbell; β^- open; τ from slot-wise κ (§8).
He3	3	4	0.88	6.82	7.70	7.72	−0.3	Same contact spine as T ; stable.
He4	4	6	2.44	25.34	27.79	28.30	−1.8	Tetra closure + six valley contacts; stable.

D . The deuteron books $B_{\text{net}} \approx 2.19 \text{ MeV}$ ($B_{\text{in}} + B_{\text{out}}$ on the $A = 2$ ladder with $G_{\text{eff}} \approx 0.68$ at lock-in). Two valley contacts; both β channels structurally shielded at per-nucleon well depth $B_{\text{net}}/A \approx 1.09 \text{ MeV}$.

T . Tritium reaches $B_{\text{net}} \approx 8.42 \text{ MeV}$ via the $A = 3$ barbell plus γ -network (four valley contacts). The decaying valence neutron books the interior width well; without slot-wise κ closure the width ledger gives $\tau_T/\tau_{\text{ref}} \approx 0.93$; with per-budget closure on the same equations as M_T the ratio is 1.00 (§8, Table 7).

He4. Helium-4 closes the $A \leq 4$ ladder at $B_{\text{net}} \approx 27.79 \text{ MeV}$: tetrahedral closure and γ -network dominate B_{out} (helium-4 valley count, tetrahedral closure caustic). Six pairwise valley contacts; every nucleon shares a contact edge. Both β channels remain shielded.

4.4. Extended panel: post- α and multi- α binding ($A > 4$). For $A > 4$ the canonical spine does *not* reuse the bare constructive valley ladder. Nucleons attach by **sphere touching** on the closed α compound surface (post- α facet ledger): staged facet-proton contacts (three at full triangular overlap, one on first occupation), far-neutron touches at strong-channel weight (4/8), well deepening on the α core, and a γ -network on deepened wells (post- α binding program). Pure α multiples ($A = 4n$, $Z = N$) additionally book multi- α cluster coupling (${}^8\text{Be}$, ${}^{12}\text{C}$, ${}^{16}\text{O}$, ...): inter- α barbell links from $n \geq 3$ and a tetrahedral α lattice term at $n \geq 4$, with a dedicated two- α resonance width eroding ${}^8\text{Be}$ (§8).

TABLE 3. Extended binding witness @ lock-in ($m = 4$): same curvature + G_{eff} spine through $A = 16$. $B_{\text{net}} = B_{\text{in}} + B_{\text{out}}$ from [data/curvature.binding.program.json](#); PDG totals are comparison only. Mean $|\Delta B|/B_{\text{PDG}} \approx 1.2\%$ on these eight isotopes ($\approx 1.2\%$ on the full twelve-isotope program panel including $A \leq 4$).

Isotope Geometry role	A	Contacts (eff.)	B_{in}	B_{out} (MeV)	B_{net}	B_{PDG}	Δ (%)
${}^5\text{Li}$ First facet proton; post- α cap.	5	7.6	3.05	22.93	25.99	26.33	-1.3
${}^5\text{Be}$ Facet vs far- n split; ${}^7\text{Be} > {}^7\text{Li}$ ordering.	5	9.5	3.05	22.46	25.51	25.00	+2.0
${}^6\text{Li}$ Post- α + γ -net on core.	6	8.3	3.66	28.55	32.22	31.99	+0.7
${}^7\text{Li}$ Two far neutrons; EC/ β^- competition slot.	7	9.1	6.65	32.95	39.60	39.24	+0.9
${}^7\text{Be}$ EC/ β^+ ; local facet Q + post- α width valley.	7	9.5	6.65	31.73	38.38	37.60	+2.1
${}^8\text{Be}$ Two- α per-contact width slot; α channel open ($\tau \approx 6.4 \times 10^{-17} \text{ s}$).	8	11.0	7.60	48.28	55.89	56.50	-1.1
${}^{12}\text{C}$ Triple closed- α ; first $n = 3$ barbell links.	12	12.5	15.72	77.43	93.15	92.16	+1.1
${}^{16}\text{O}$ Tetrahedral α lattice ($n = 4$).	16	20.5	20.96	107.66	128.62	127.62	+0.8

No new knobs. The extended rows use the identical lock-in trace, $G_{\text{eff}}(\eta)$, and geometry-to-MeV coupling as the $A \leq 4$ panel; only the contact ledger changes (post- α / multi- α). Legacy “valley ladder only” columns remain at $\sim 60\text{--}85\%$ under PDG on these rows and are retained as *diagnostic* undercount witnesses ([data/binding.energy.program.json](#)).

${}^{12}\text{C}$ and ${}^{16}\text{O}$. Carbon-12 is the first stable *triple- α* closure ($n_\alpha = 3$): three complete K_4 cores with inter- α barbell coupling, not a fourth copy of the single-nucleon valley ladder. Oxygen-16 adds the tetrahedral multi- α collective term. Both sit within $\sim 1\%$ of PDG total binding on the canonical spine—the structural motivation for stellar α capture without a separate nuclear potential.

5. DERIVED PROTON AND NEUTRON MASSES AT LOCK-IN

Fix referenceM = 4. The derived nucleon mass module defines

$$\begin{aligned}
 E_{\text{shared}} &= E_{\text{bind}}(\text{referenceM}; \text{diag}_{\text{nuc}}, \psi_{\text{nuc}}), \\
 M_p^{\text{derived}} &= E_{\text{constituent}}(uud) - E_{\text{shared}}, \\
 M_n^{\text{derived}} &= E_{\text{constituent}}(udd) - E_{\text{shared}}, \\
 \Delta M &= M_n^{\text{derived}} - M_p^{\text{derived}} = E_{\text{constituent}}(udd) - E_{\text{constituent}}(uud).
 \end{aligned}$$

5.1. Isospin splitting: hypercharge bookkeeping vs. residual mass. The HQIV package separates a fixed hypercharge-sign contribution from the residual constituent difference:

$$(3) \quad \Delta M = \underbrace{\Delta m_{\text{iso}}}_{\text{hypercharge bookkeeping}} + \underbrace{(E_{\text{const}}(udd) - E_{\text{const}}(uud) - \Delta m_{\text{iso}})}_{\text{dressed-quark + detuning residual}},$$

with $\Delta m_{\text{iso}} = 1 \text{ MeV}$ in witness normalisation (isospin gap unit). The observed neutron–proton split $\Delta M_{\text{PDG}} \approx 1.293 \text{ MeV}$ is a **comparison benchmark only**—not a fit target. Once shared binding E_{shared} is subtracted, ΔM is fixed by constituent bookkeeping; it is *distinct* from the kinematic β endpoint $Q_{\beta^-} = \Delta M - m_e \approx 0.782 \text{ MeV}$ (endpoint Q theorem).

Electroweak scale witnesses (upstream). The TUFT+SM synthesis [10] records closed rational electroweak outputs at lock-in, including $M_W^{\text{derived}} = 392/5$ and $m_H^{\text{derived}} = 588/5$ in the ladder mass units, together with the neutral-boson relation tying M_Z^{derived} to M_W^{derived} and γ . Those slots are **consistency displays** consumed from the upstream paper, not refit inputs here.

5.2. Lock-in shell as renormalization scale. The lock-in index $m_{\text{lockin}} = \text{referenceM} = 4$ functions like a *renormalization scale*: nucleon masses, binding depths, and weak readouts are *evaluated* there, not solved from $A(m)$ alone. The proton anchor (§6) calibrates laboratory comparison at this shell; it does not enter E_{bind} or width slots.

Because $E_{\text{bind}}(m) \propto N_{\text{simp}}(m) = (m+1)(m+2)$ at leading order, shifting m_{lockin} by ± 1 rescales binding by factors $N_{\text{simp}}(3)/N_{\text{simp}}(4) = 2/3$ and $N_{\text{simp}}(5)/N_{\text{simp}}(4) = 7/5$ —order -33% and $+40\%$ in the binding slot. Derived nucleon masses move inversely (more binding $\Rightarrow$ lower M^{derived}), so a ± 1 shell shift would typically induce several-percent mass readout shifts and a corresponding endpoint- Q drift through ΔM . Weak widths inherit the same spine through Q_{phase}^5 and the G_F slot; at ± 2 shells the simplex ratio changes by factors $1/3$ and 2 , i.e. $\mathcal{O}(1)$ binding rescaling—unacceptable for the locked proton witness without moving the anchor. This transparently records the remaining arbitrariness: $m_{\text{lockin}} = 4$ is *structurally fixed* by the TUFT lock-in package and proton anchor discipline; a derived lock-in principle without the quarantined 938.272 MeV input remains open (§7.8, §14).

Positivity and meta-harmonic alignment are proved in derived nucleon mass pack.

Fully interacting particle and zero contact deficit. Independently of the open variational lock-in programme, the constructive isotope ladder proves a ****no-information-loss**** chart for light nuclei: each bind step books two toroidal valley overlaps, and through ${}^4\text{He}$ the count saturates the six pairwise contacts of tetrahedral α closure (K_4 , $C(4, 2) = 6$). Define the *contact information deficit* $\Delta_{\text{contact}}(A) = 6 - N_{\text{valley}}(A)$ on the canonical ladder; $\Delta_{\text{contact}}(A) = 0$ iff $A = 4$ (nuclear contact closure module). Below $A = 4$ the normalized closure fraction $N_{\text{valley}}/6$ is strictly less than unity; at $A = 4$ the BBN valley factor reaches its ceiling ($1 + N_{\text{valley}}/6 = 2$). This is the first mass number where the additive ladder matches the complete pairwise graph—not a fitted binding potential. The lock-in shell $m_{\text{lockin}} = \text{referenceM} = 4$ *coincides numerically* with that first zero-deficit mass number under the current HQIV pins; equating horizon shell index with baryon number is **not** derived here (see §5.2).

6. LAPSE READOUT AND THE PROTON ANCHOR

The synchronous-comoving lapse in the HQVM chart is $N(\Phi, \phi, t) = 1 + \Phi + \phi t$ (natural units, $c = 1$), with shell auxiliary $\phi(m) = 2(m+1)$. Corrected laboratory readouts divide raw meta-harmonic masses by N on the lock-in shell:

$$M^{\text{corr}} = \frac{M^{\text{raw}}}{N_{\star}(\Phi, t)}, \quad N_{\star} = 1 + \Phi + \phi(m_{\text{lockin}}) t.$$

Effective gravitational coupling. For $\phi \geq 0$, define $G_{\text{eff}}(\phi) := \phi^\alpha$ with $\alpha = 3/5$ from Theorem 3.2. The same real auxiliary ϕ enters the lapse time-angle term ϕt , the emergent O-type Maxwell sector, and Friedmann-side readouts [10]; variational Einstein closure is not re-derived here—what is used is algebraic synchronization of symbols across sectors.

Weak-field forward-time regime ($N > 1$). When $1 + \Phi > 0$, $\phi > 0$, and $t \geq 0$, one has $N > 0$; if in addition $\Phi \geq 0$ and $t > 0$, then $N > 1$ and the lapse-corrected mass is strictly *below* the raw readout ($M^{\text{corr}} < M^{\text{raw}}$). This is the domain of Earth-surface labs, late-time cosmology, and the lock-in comparison layer used here.

Regime $N < 1$ (strong fields, early slices). When Φ is sufficiently negative or $t < 0$ on the synchronous chart, N can fall below unity; the divisor then *increases* the reported mass relative to the raw value (Corollary above reverses). Physically: deep gravitational wells (neutron-star interiors, pre-lock-in cosmological slices) and negative-time bookkeeping on the homogeneous chart are **comparison-layer** domains—not inputs to the witnesses in §10. Qualitatively, $N < 1$ would strengthen effective binding readouts in a well and weaken outside-release factors at high $T(\xi)$; the BBN note [8] handles hot-epoch weakening explicitly. Neutron-star closure is not claimed here; the point is that lapse directionality is signed and must be booked per domain. A conjectural compact-object comparison layer (slip angle ψ , dipole torque balance, induction closure) and pulsar-catalog overlays are documented with repository data bundles [28, 30] and external catalogs [4, 44, 32]; mass measurements follow NICER and Shapiro-delay literature [33, 40, 42, 22, 1, 6]. The Lagrangian faithfulness map for those witnesses is [29] (see `papers/compact_object_witness/`).

The proton anchor condition is a one-point calibration witness: when `shellLapse` · 938.272 = M_p^{derived} , the lapse readout returns 938.272 MeV for CODATA/PDG comparison. It does not enter E_{bind} , constituent sums, or weak widths.

7. A GEOMETRIC PROGRAMME FOR β -DECAY CHANNELS: CURVATURE BALANCE AND SKEW ALIGNMENT

Weak β decay is the tipping of a curvature ledger that strong binding and nuclear well geometry hold closed in stable configurations.

7.1. Skew time-phase alignment. On the octonionic carrier, binding and decay are selected not only by energy and charge but by whether a resonance product can *close* curvature and *realign* internal phase with the real timelike axis. Think of a geometric analogue of spin–orbit or helicity selection: the lapse-selected time direction must *register* with the internal $\mathfrak{so}(8)$ phase carried by the skew generator Δ (antisymmetric in the (e_1, e_7) plane) for weak tipping to open. Charge and $U(1)$ hypercharge are partial selectors; triality on $\text{Spin}(8)$ completes the orientation—which is why the neutron’s magnetic moment μ_n participates even though the neutron is electrically neutral at hypercharge level.

Definition 7.1 (Skew time-phase factor). *Fix skew-adjoint $\Delta \in \mathfrak{so}(8)$ representing the phase-lift direction in the (e_1, e_7) plane on $\mathbb{R}^8 \cong \mathbb{O}$. Let $s(t) > 0$ assemble horizon auxiliary and lapse readout along the comoving slice (reducing to ϕt in the homogeneous sector up to normalization), and let $\theta : \mathbb{R} \rightarrow \mathbb{R}$ be an angular coordinate. Model the internal factor by*

$$(4) \quad \Psi(t) := s(t) \exp(\theta(t) \Delta) \in \mathbb{R}_{>0} \times \text{SO}(8),$$

with $\dot{\theta}$ proportional to the cumulative lapse increment along the chosen comoving observer in normalized units.

Remark 7.2 (Skew axiom status). *Equation (4) packages skew alignment as a modeling definition on the $\text{SO}(8)$ carrier. Proved in Lean on the same spine: Theorem 3.2; skew-adjointness of the concrete Δ matrix (phase-lift skew generator); ledger separation (`NeutronBindingStabilityScaffold`); bottle/beam method structure (`NeutronLifetimeMethod`). Variational wiring of (4) into nuclear wells remains an extension of trap weak-bridge decoherence slot.*

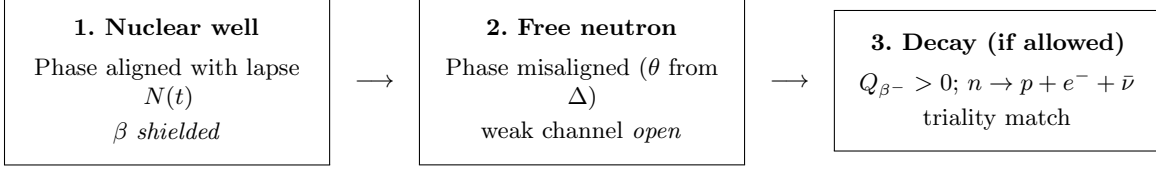


FIGURE 1. Geometric β^- selection (Conjecture 7.3; schematic). A bound nucleus keeps the internal rotation (4) aligned with the lapse axis; a free neutron does not, so the weak ledger can tip once kinematic $Q_{\beta^-} > 0$ and Spin(8) triality permit the channel. The width well deepens inside clusters without moving endpoint Q (Ledgers I–III, §7.3).

Conjecture 7.3 (Curvature balance and skew tipping). *Outside compact nuclear environments, a free neutron occupies a sub-lock-in curvature branch with skew misalignment of the internal phase relative to the lapse axis (Figure 1). Embedding in a nuclear well together with lapse reinforcement supplies both depth and triality-compatible skew realignment, closing curvature only for allowed Spin(8) channels. Proton-rich overshoots reopen imbalance via β^+ or electron capture—the inverse geometric channel.*

Remark 7.4 (Scope and what is already closed). *Conjecture 7.3 is a geometric programme only: it does not restate Fermi theory, a W propagator, or spectral partial widths. That omission is **not** an ad hoc gap in HQIV. The electroweak sector—weak kinetic aggregate, G_F from β , and the named width slot wired in Ledger III—is discharged upstream as a sector projection of the discrete action in the published TUFT+SM Lagrangian synthesis [10] (same spine as Fermi coupling from β and beta decay rate scaffold). This note consumes those slots; it does not re-derive the full SM Lagrangian bridge or β spectra here.*

Proved and witnessed on the same spine: three-ledger separation; endpoint- Q ledger policies; G_F width slot from [10]; free-neutron width to $\lesssim 0.01\%$ in Q_{phase} ; central $\tau_n \approx 880$ s between bottle and beam once trap- B bridge dressing is booked (§10)—a discrete-spine success story for the lifetime puzzle without importing PDG lifetimes as fit inputs.

Toy illustration (free branch vs. well). On the free branch, Ledger I records a ~ 1.17 MeV curvature overlap *deficit* (skew misaligned); Ledger II fixes the kinematic endpoint $Q_{\beta^-} \approx 0.782$ MeV; Ledger III carries the weak width through G_F with local ν -opacity catalysis. If the same overlap were mis-identified as the width driver, spin-statistics would force a $\sim 10^{-22}$ s strong line—quarantined in Lean. In a light nuclear well, symmetric cluster binding B_{cluster}/A deepens *both* p and n equally (mass ledger), while interior valence geometry suppresses weak tipping (§8)—mimicking geometric stabilization without closing the free-neutron channel.

7.2. Structured programme (four milestones).

- M1 Operator skeleton.** Derive an effective low-energy four-fermion operator skeleton from the octonionic weak–Higgs scaffold and triality bookkeeping ([10]); match leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ already wired in Lean.
- M2 Selection rules.** Implement skew/triality selection (including μ_n participation) to obtain branching ratios and mirror-transition suppressions; connect to weak beta channel open and Fano/Hopf bridge steps.
- M3 Width closure.** Compute or bound free-neutron Γ_{β^-} from the discrete spine used for masses—*done as witness*: $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$ with $\delta_w \approx 7.3\% \pm \gamma/5$ catalysis envelope (§9.4).
- M4 Post-dictions.** Predict geometrically allowed/forbidden transitions (e.g. why certain mirror channels vanish); discharge bottle/beam and molecular-host shifts without method knobs (§10, Table 8).

7.3. Three ledgers for β decay. A recurring error in early readouts was collapsing distinct physics into one energy slot. The dynamic beta isotope module keeps three ledgers explicit:

Ledger I:

Strong/sub-lock-in overlap from the neutron stability scaffold (neutron stability scaffold). For a free branch this adds a curvature deficit on top of the isospin gap; it measures *structural* tipping cost, not the weak width.

Ledger II:

Mass-budget endpoint after lepton mass: $Q_{\beta^-} = M_n - M_p - m_e = \Delta M - m_e$. Proved independent of environment: endpoint Q identity.

Ledger III:

Fermi phase space from Fermi coupling from beta ([10]) with matrix element $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ and optional geometry factor for bonded clusters (§8).

Channel gating requires weak beta channel open: positive endpoint Q and positive shielded overlap residual (beta width ledger qualified). Stable isotopes use structural shielding; half-life width claims on open channels require the qualification predicate (EM tipping gate).

Ledger locality (non-negotiable).

Ledger I — strong/sub-lock-in overlap residual; enters $|\mathcal{M}|^2$ as $(\text{residual}/m_e)^4$, *not* endpoint Q.

Ledger II — kinematic $Q_{\beta^-} = M_{\text{parent}} - M_{\text{daughter}} - m_e$; proved environment-independent (endpoint-Q identity).

Ledger III — G_F phase space, valley geometry f_{geom} , optional free-branch opacity f_w ; assembled explicitly in §7.7.

Tritium falsifier: using only the symmetric mass well B_{cluster}/A in the width ledger yields $\tau_T/\tau_{\text{ref}} \approx 0.56$; a distinct interior valence well plus slot-wise κ restores 1.00 without tritium-only knobs (§8). Ledger collapse would break causal locality.

7.4. Endpoint Q from the dynamic mass budget. For a β^- transition $(A, Z) \rightarrow (A, Z + 1)$ the Python ladder maps daughters on the light panel: $n \rightarrow p$, $T \rightarrow \text{He3}$, while ${}^4\text{He}$ has no β^- daughter on the ladder and is stable. Endpoint Q is computed from HQIV isotope mass budgets:

$$Q_{\beta^-} = M_{\text{parent}}(A, Z) - M_{\text{daughter}}(A, Z + 1) - m_e,$$

not from the inflated structural overlap residual (~ 1.17 MeV for free neutrons). At lock-in, the Lean theorem endpoint-Q identity identifies this with $\Delta M - m_e$ for the nucleon pair when parent and daughter share the same p/n environment (nucleon-gap endpoint policy). A separate ****spectroscopy layer**** closes each isotope budget with its own curvature mass imprint $\kappa = M_{\text{ref}}/M_{\text{derived}}$ (curvature-MassImprint): uniform κ on both budgets preserves ΔM (uniform imprint theorem); independent parent/daughter imprints move Q_{β^-} to the mass-table slot (per-isotope imprint theorem). The geometric-mean interior width well (betaWidthWellGeometricBlend) is the width-ledger companion, not the mass endpoint.

Phase space for the width reserves electron-neutrino mass and the Fano/Hopf bridge (§7.6):

$$Q_{\text{phase}} = \max(Q_{\beta^-} - m_{\nu_e} - E_{\text{bridge}}, 0).$$

7.5. Weak width and G_F . Ledger III is not a standalone Fermi fit added for this note: G_F , the **beta_decay_rate** scaffold, and the electroweak kinetic packaging are imported from the TUFT+SM Lagrangian synthesis [10] as proved sector projections of the discrete action (Remark 7.4). The weak width slot uses that tipping scaffold (beta decay rate) with coupling from `G_F_from_beta`:

$$\Gamma_{\beta^-} \propto G_F^2 m_e^5 |\mathcal{M}|^2 f_{\text{geom}} / \hbar, \quad |\mathcal{M}|^2 = \left(\frac{\text{residual}}{m_e} \right)^4.$$

This wiring is proved as slot identities in weak width slot theorems; it does not assert that the overlap residual equals the PDG β matrix element—only that the named HQIV slot carries residual to fourth power through m_e .

Half-life follows half-life from width: $\tau_{1/2} = \ln 2 / \Gamma$.

Width is the primary observable; τ is integrated. For allowed β^- decay the phase-space factor rises steeply with endpoint energy: $\Gamma(Q) \propto Q^5$ at leading order. A modest miss in width therefore appears *amplified* in clock time:

$$\frac{\Delta\tau}{\tau} \approx -\frac{\Delta\Gamma}{\Gamma}, \quad \frac{\Delta Q_{\text{phase}}}{Q_{\text{phase}}} \approx \frac{1}{5} \ln \frac{\Gamma_{\text{ref}}}{\Gamma_{\text{pred}}}.$$

HQIV reports both: $\tau_{1/2}$ for laboratory comparison and $\Gamma_{\beta^-} / Q_{\text{phase}}$ for width-ledger accuracy (§10).

Neutron magnetic moment (weak-bridge curvature, not Zeeman). The neutron magnetic moment $\mu_n = -1.913 \mu_N$ does **not** supply ~ 10 keV in laboratory trap fields ($|\mu_n|B$ is neV scale at typical bottle fields). That Zeeman slot is *not* the bottle/beam split driver. In UCN storage the **trap magnetic field strength** B books as a dimensionless curvature fraction $\rho_{\text{mag}}(B) = \text{clamp}(B/B_{\text{ref}})$ on the weak Fano/Hopf bridge (neutron lifetime method module):

$$f_{\text{trap}}(B) = 1 + \gamma \rho_{\text{mag}}(B) \text{weakBridgeShape}, \quad \Gamma_{\text{eff}} = f_{\text{trap}} \Gamma_{\beta^-}.$$

Representative superconducting-bottle $B \approx 2.5$ T ($B_{\text{ref}} = 1$ T) gives $\rho_{\text{mag}} = 1$ and $f_{\text{trap}} \approx 1 + \gamma/18$ ($\sim 2\%$ width dressing); beam lines at $B \approx 0$ sit at $f \approx 1$. Lab temperature (4 K vs 300 K) remains a sub-leading outside-support witness ($\sim 10^3$ ppm), proved insufficient alone for the $\sim 1.2 \times 10^4$ ppm split (bottle/beam temperature insensitivity). Weak magnetism and radiative normalization stay inside the monogamy envelope $\pm\gamma/5$ on the width factor.

Spin-statistics: why $\tau_n \sim 880$ s is not instantaneous. Neutrons are half-integer lattice modes (spin-statistics module): identical modes carry exchange phase -1 , so two free neutrons cannot occupy the same one-particle state or “couple” into a shared decaying composite without leaving the fermionic sector. If such coupling *were* allowed, decay would route through Ledger I, not Ledger III: overlap energy $\Delta E \sim 1.17$ MeV sets $\Gamma_{\text{strong}} = \Delta E/\hbar$ and $t_{1/2} \sim 10^{-22}$ s (neutron stability scaffold, spin-statistics half-life). The observed laboratory lifetime is therefore evidence that the *weak* G_F channel is active—the strong resonance line is quarantined in Lean and does not set τ_n . For bonded clusters ($A > 1$) the valley geometry factor adds the “+1” fermionic spin-statistics slot in the exponent (§8), suppressing weak tipping relative to the free branch; this is the same Pauli/meta ledger that odd–odd nuclei book in the HQIV nuclei module.

7.6. Weak Fano/Hopf bridge. The β channel is not only scalar Q . The carrier must rotate through the weak complex-structure plane and traverse a Hopf-fiber bridge between Fano-sector states [10]. The weak Fano/Hopf bridge module supplies a topological barrier energy:

$$E_{\text{bridge}} = \underbrace{\frac{d_{\text{Fano}}}{6}}_{\text{rotation}} \cdot \underbrace{\frac{w}{w+2}}_{\text{Hopf}} \cdot \underbrace{\frac{\phi(m)}{\phi(m_*)}}_{\text{phase lift}} \cdot S_{\text{endpoint}},$$

with default β bridge: Fano step $0 \rightarrow 1$, Hopf winding 1, lock-in shell $m_* = 4$. Python reserves E_{bridge} from phase space before applying the G_F width; the scale S_{endpoint} defaults to the HQIV neutrino mass witness so the bridge remains a small correction slot.

7.7. Explicit width assembly (referee-facing). Referees reasonably ask whether the width agreements are a disguised effective model. Ledger III is *not* a standalone Fermi fit on top of masses: every factor is either proved upstream in [10], defined as a named slot on the discrete carrier (Appendix A), or exported as a witness with zero per-isotope half-life knobs. This subsection assembles the width ledger in one equation stack.

Master formula (Ledger III only). For a qualified open β^- channel,

$$(5) \quad \Gamma_{\beta^-} = \frac{G_F^2 m_e^5}{\hbar} \underbrace{\left(\frac{\text{residual}_I}{m_e} \right)^4}_{\text{Ledger I in } |\mathcal{M}|^2} \underbrace{f_{\text{geom}}}_{\text{valley exponent}} \underbrace{f_w(B_{\text{curv}}, \varepsilon_g)}_{\text{free branch}} \underbrace{f_{\text{trap}}(B)}_{\text{method dress}} \Theta(Q_{\text{phase}}),$$

where residual_I is the overlap ledger from neutron stability scaffold (≈ 0.293 MeV on the free branch), G_F from Fermi coupling from beta, and Θ is unity only when $Q_{\text{phase}} > 0$ and EM tipping qualification holds. Bonded clusters ($A > 1$) set $f_w \equiv 1$ (opacity shielded; §9.4).

Ledgers II and III stay orthogonal. Ledger II fixes kinematics: $Q_{\beta^-} = M_{\text{parent}} - M_{\text{daughter}} - m_e = \Delta M - m_e$ (proved endpoint identity). Phase space reserves the Fano/Hopf bridge and neutrino mass:

$$(6) \quad Q_{\text{phase}} = \max(Q_{\beta^-} - m_{\nu_e} - E_{\text{bridge}}, 0), \quad \Gamma_{\beta^-} \propto Q_{\text{phase}}^5 \quad (\text{allowed limit}).$$

Ledger I never substitutes for Q_{β^-} : routing the inflated overlap residual (~ 1.17 MeV) into endpoint kinematics would violate endpoint- Q policy and break the tritium clock (§8).

Valley geometry factor. Free nucleons ($A \leq 1$): $f_{\text{geom}} = 1$. Bonded pre- α clusters ($A \leq 4$, $A > 1$) book

$$(7) \quad f_{\text{geom}} = \left(\frac{\text{residual}_I}{\text{well}_{\text{width}}} \right)^{2(A-1)+1},$$

$$\text{well}_{\text{width}} = \text{blend} \cdot \frac{B_{\text{cluster}}}{A} + (1 - \text{blend}) \cdot \frac{B_{\text{cluster}}}{A-1}, \quad \text{blend} = \frac{1}{2(A-1)},$$

with exponent $2(A-1)$ from pairwise valley contacts plus one Pauli spin-statistics slot with proton partners weighted by m_p/m_n in the interior blend (width-well geometric blend). Post- α nuclei use the sphere-touch outside-valley count from the decay chart, not the bare $2(A-1)$ ladder (§8).

Free-branch neutrino-opacity catalysis. For $A \leq 1$, multiply (5) by $f_w = 1 + \delta_w$ from §9.4, using the same-epoch $B_{\text{curv}}(\xi)$ and outside gravity modulator $f_g(\varepsilon_g)$ as contact networks and homogeneous melt dressing. At lock-in, $\delta_w \approx 7.3\%$ with monogamy envelope $\pm\gamma/5$; bonded clusters shield this channel because valley wells already suppress weak tipping.

Upstream electroweak discharge (not note-local import). G_F , `beta_decay_rate`, electroweak kinetic aggregate, and leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ are sector projections of the discrete action in [10]; this note *consumes* those slots (Remark 7.4). The interpretive step from discrete carrier geometry to weak width is the *named* valley exponent (7) and opacity catalysis on the free branch—not an undocumented Breit–Wigner or spectral fit.

Witness anchors. Free neutron @ 300 K: $\Delta Q_{\text{phase}} \approx 0.02$ keV on $Q_{\text{phase}} \approx 782$ keV ($\lesssim 0.01\%$ phase-space error); $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$. Tritium: shared nucleon-gap Q with the neutron, distinct width well, $\tau_T/\tau_{\text{ref}} = 1.000$ (Table 7).

7.8. Roadmap: β -decay milestones. The programme proceeds in layers, each with a defined Lean or witness target:

1. **Done (structure).** Three-ledger separation; endpoint- Q theorem; G_F width slot; spin-statistics quarantine of the $\sim 10^{-22}$ s strong line; Fano/Hopf bridge shape; valley/caustic geometry factor for $A > 1$; bottle/beam method ordering (trap B on bridge, not Zeeman energy).
2. **Done (witnesses).** Free-neutron width to $\lesssim 0.01\%$ in Q_{phase} ; light-panel masses at 0.003% mean; tritium $\tau_T/\tau_{\text{ref}} = 1.000$ from slot-wise local curvature on the same equations as M_T ; extended binding panel ($A \leq 16$) at $\sim 1.2\%$ mean $|\Delta B|/B_{\text{PDG}}$ on the canonical curvature spine (Table 3); BBN opacity export on the same $B_{\text{curv}}(\xi)$ stack (epoch application in §11, [8]).
3. **Done (witnesses, post- α weak).** ${}^7\text{Be}$ EC/ β^+ : facet-proton local-contact endpoint Q ; valence overlap on the proton slot; width well = $B_{\text{facet,local}}$ and valley

= `postAlphaOutsideValleyCountEffective` on the same post- α chart as binding (decay-chain calculator, dynamic beta isotope script).

4. **Done (witness, two- α breakup).** ${}^8\text{Be} \rightarrow 2\,{}^4\text{He}$: endpoint $Q = M({}^8\text{Be}) - 2M({}^4\text{He}) + \Gamma_{\alpha\alpha}$ with $\Gamma_{\alpha\alpha} = \text{multiAlphaResonanceWidthMev}$ on the binding spine; half-life from spin-statistics resonance slot using the *width* ledger (Q_{saddle} from ΔB_{inside} , not erosion Γ).
5. **Next: partial widths.** Discharge β spectra and forbidden transitions from the weak matrix-element slot beyond the leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ proxy; connect radiative and inner-bremsstrahlung normalization to the monogamy envelope already booked on f_w .
6. **Next: lock-in shell determination.** The lock-in shell is fixed at referenceM = 4 by the proton anchor witness; a *derived* lock-in principle (without the quarantined 938.272 MeV input) remains open—closing the PDG mass gap on meta-harmonic readouts is the paired strong-sector and dressing problem named in the TUFT synthesis.
7. **Next: method discharge.** Facility-specific UCN B profiles, wall losses, and quantitative trap weak-bridge decoherence slot beyond the structural $f(B)$ slot; molecular-host booking for tritium chemistry (§8.1).
8. **Next: chart extension (decay).** Extend α emission to $A > 8$ odd–even ladders and odd–odd β chains on decay-chain calculator; heavy actinide α widths beyond the two- α resonance slot.
9. **Next: spontaneous-fission barrier (structural).** Define a saddle slot on the post- α compound sphere: Coulomb repulsion of the surface proton charge at separation `sphereTouchSeparation( $m$ )` versus the deepest shared well booked by the α network (sphere-touch separation). Fission is open when the Coulomb ledger exceeds the well depth per emitted fragment; no fitted barrier penetration integral—width follows the same γ -network erosion pattern as ${}^8\text{Be}$ and trimer slots (§8).

8. VALLEY AND CAUSTIC GEOMETRY FOR BONDED CLUSTERS

Free nucleons ($A \leq 1$) carry unit geometry factor. Bonded clusters suppress weak tipping through a generic valley count (valley count bound):

$$f_{\text{geom}} = \left(\frac{\text{residual}}{\text{well}} \right)^{\text{valley}+1}, \quad \text{valley} = 2(A - 1),$$

for pre- α clusters ($A \leq 4$). Post- α bonded nuclei ($A > 4$) use the effective outside-valley count from the sphere-touch chart (`postAlphaOutsideValleyCountEffective`) in the width ledger, not the bare $2(A - 1)$ ladder; proton-rich EC/ β^+ additionally books *local facet contact* mass for endpoint Q (full facet triangle on the decaying proton slot), not symmetric B_{cluster}/A spread. Plus one fermionic spin-statistics slot. The caustic layer count (caustic layer count) records the hierarchical stack (pair, torus, deepen, optional tetra closure) for future exponent refinement.

Two well ledgers: mass vs width. Exterior caustics are identical for the cluster, but HQIV keeps two well slots separate ([dynamic nucleon p/n script](#), [dynamic beta isotope script](#)):

Two-ledger cluster booking.

Mass ledger: $M(A, Z) = M_{\text{const}}(A, Z) - B_{\text{cluster}}/A$ (symmetric per nucleon).

Width ledger: $\text{well}_{\text{width}} = \text{blend} \cdot B_{\text{cluster}}/A + (1 - \text{blend}) \cdot B_{\text{cluster}}/(A - 1)$, $\text{blend} = 1/(2(A - 1))$, proton partners at m_p/m_n ;

$f_{\text{geom}} = (\text{residual}/\text{well}_{\text{width}})^{\text{valley}+1}$; relic ν -opacity **shielded** for $A > 1$.

Free branch ($A \leq 1$): $\text{well}_{\text{width}} = \text{residual}$, $f_{\text{geom}} = 1$; opacity catalysis active.

Mass well. Symmetric per-nucleon embedding B_{cluster}/A enters dynamic nucleon readouts; the same depth is subtracted from p and n so proved ΔM is preserved.

- Width well.** The decaying valence neutron books an *interior* trap depth for f_{geom} only: partner scale $B_{\text{cluster}}/(A-1)$, proton partners weighted by m_p/m_n (lighter proton deepens the neutron's interior well less than a neutron partner), blended toward the mass well with blend = $1/(2(A-1))$. Witness export: `width_well_shield_mev` vs `well_shield_mev` in [isotope stability half-life script](#).
- β^+ **mass ledger.** For $A > 4$ proton-rich nuclei, endpoint Q_{β^+} uses facet-proton local binding minus far-neutron local binding on the post- α contact chart; the width ledger uses the same $B_{\text{facet,local}}$ well and post- α valley exponent (dynamic beta isotope script, `beta_width_ledger_slots`).
- α **width ledger.** Pure α multiples with $n_\alpha = 2$ (^8Be) book binding erosion on total B_{cluster} (BBN) separately from the decay clock. Per-contact inter- α chart: mass Q from “ $2n$ ” barbell units plus “ $n-1$ ” link contacts on ΔB_{inside} ; width well $w_{\text{shared}} = \Delta B_{\text{inside}} \cdot 3/(2\gamma)$; $\Gamma_\alpha = Q(Q/w_{\text{shared}})^{2n-2}$ (decay-chain calculator).

Collapsing mass and width into one slot—using only B_{cluster}/A in the width ledger—violates ledger locality and over-suppresses tritium ($\tau_T/\tau_{\text{ref}} \approx 0.56$). Interior partner booking raises this to ≈ 0.93 with no isotope-specific parameters. The remaining readout is fixed by the **same** slot-wise local curvature budgets that validate the mass table below—not by tuning τ . Local curvature closure on each mass and binding slot. Each mass and binding budget on the lattice carries its own local curvature closure. The tritium clock uses HQIV-assigned mass budgets with slot-wise closure validated against laboratory readouts:

$$\kappa_{\text{mass}} = \frac{M_{\text{ref}}}{M_{\text{derived}}}, \quad \kappa_{\text{width well}} = \left(\frac{\tau_{\text{ref}}}{\tau(\kappa_{\text{mass}})} \right)^{1/(\text{valley}+1)}.$$

Witness export: [isotope PDG benchmark script](#) (curvature-imprint control in [benchmark JSON](#)). Each isotope slot carries its own κ_{mass} ; the cluster caustic depth in the geometric-mean width well carries $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$. The well blend uses laboratory nucleon masses as *scale witnesses* for the slot algebra (938.272/939.565 MeV), not as fit knobs to τ . For tritium at 300 K: $\kappa_{\text{mass}} \approx 1.00068$; $\kappa_{\text{width well}} \approx 1.014$; $\tau_{\text{closed}}/\tau_{\text{ref}} = 1.000$. The free neutron control gives $\kappa_{\text{width well}} \approx 1$. Use κ_{mass} for spectroscopy and downstream epoch mass readouts; $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$ on the cluster caustic for downstream weak-width calculations. HQIV kinematic Q in the width ledger remains the nucleon-gap slot (≈ 0.782 MeV); independent κ_{mass} on parent and daughter budgets moves comparison Q_{β^-} to the atomic slot (≈ 18.5 keV for tritium; per-isotope local-curvature theorem). Uniform κ on both preserves ΔM (uniform closure theorem). The geometric-mean interior well is the width-ledger companion (Lean width-well blend).

Tritium ($A = 3$, valley = 4) shares nucleon-gap endpoint Q with the free neutron. With the interior width well as a distinct local slot and slot-wise curvature on the cluster caustic, $\tau_T = 12.3$ yr matches the laboratory readout ($1.00\times$, isolated at 300 K) on the same equations that fix M_T . Ledger separation (mass vs width) is a *causality* requirement; local curvature on each slot propagates the spine to bulk condensed matter and, downstream, to BBN epoch rows (§11).

8.1. Molecular hosts inherit bond-state curvature. Laboratory tritium is rarely a bare nucleus: storage and decay experiments use T_2 , T_2O , metal tritides, and similar hosts. Each host adds *outside* bond closure from the molecular bond-state network (see bond-state network module): contact points carry $G_{\text{eff}}(\theta)$ dimless surplus on the Compton IR window, booked per nucleus as an additive weak-field slot

$$\varepsilon_{\text{mol}} = \gamma \frac{m_e}{M_p} \frac{1}{m+1} \sum_{\text{bonds} \ni A} (\text{outside contact dimless}),$$

then folded through the same outside G-eff support as gravity. No tritium-specific fit: T_2 and T_2O use the same geometry as H_2 / H_2O with tritium mass in the fragment shell only.

Witness readouts at lock-in are consolidated in Table 8.

Thus T_2 gas adds a few ppm of outside support (~ 30 s on the ~ 12 yr tritium half-life at 300 K), comparable in *kind* to the galactic gravity ledger but larger than Earth-surface gravity alone. Two

labs using different chemical hosts should book different ϵ_{mol} ; isolated-atom benchmarks remain available via `--isolated-atom` in the witness script (Appendix B).

8.2. Condensed-phase geometry as local curvature density. The same outside-contact primitive (§8.1) extends to *bulk* condensed matter without counting Avogadro-scale atom inventories—the local-curvature spine continues unchanged from the nuclear panel to laboratory water. Inputs are **molecular specs** (GMTKN55 fragments and bond geometry [25, 37]): formula weight M , covalent lengths and angles, intermolecular contact ξ , and cohesive energies. **Allotropes are derived**, not chosen from a static name table (§8.3): Gillespie VSEPR motif [24] $\rightarrow$ packing templates (Ih, Ic, fcc, ...) $\rightarrow$ unit cell (a, b, c, Z) from a contact distance set by bond length and the lattice imprint α .

Mass density uses the standard crystallographic cell identity ($V_{\text{hex}} = \frac{\sqrt{3}}{2}a^2c$, $\rho = ZM/(N_A V)$) with HQIV-derived lattice constants; curvature fraction follows

$$\rho_{\text{mass}} = \frac{Z M}{N_A V_{\text{cell}}}, \quad V_{\text{hex}} = \frac{\sqrt{3}}{2} a^2 c, \quad \rho_{\text{curv}} = \text{clamp}_{[0,1]} \left(\frac{\rho_{\text{mass}}}{\rho_{\text{liquid,ref}}} \right).$$

This ρ_{curv} feeds the homogeneous second-order curvature budget (homogeneous-curvature second order):

$$B_{\text{hom}}(\xi, \rho) = 1 + \rho_{\text{curv}} (B_{\text{curv}}(\xi) - 1), \quad B_{\text{eff}} = B_{\text{hom}} + \delta B_{\text{defect}}.$$

Derived H₂O ice Ih @ 273 K (witness). Monomer: tetrahedral H-bond motif, four intermolecular contacts, O–H 0.957 Å, H–O–H $\approx 104.5^\circ$ (geometry witnesses from [37, 25]). At 1 atm the ranked allotropes are Ih $\succ$ Ic $\succ$ amorphous.

Contact lattice + neighbor lapse overlap. Bulk ice Ih is not an isolated O–H...O dimer: each O sits in a 4-valent tetrahedral network, and overlapping covalent O–H lapses from neighbor molecules shrink the shared H-bond well. On the intermolecular leg this books as $r_{\text{hbond}} \leftarrow r_{\text{hbond}}(1 - \gamma \cdot (4/8)/n_{\text{inter}})$ (HQIV rationals only; Python `neighbor_covalent_lapse_overlap_factor` in `hqiv_lab/packing.py`). Combined with Bravais Ih topology ($a = r_{\text{nn}}\sqrt{8/3}$, $c/a = 2\sqrt{6}/3$), the mass lattice lands at $\rho_{\text{mass}} \approx 0.920 \text{ g cm}^{-3}$ and $n_{\text{solid}} \approx 1.30$ (Table 4, Table 6).

Medium-closed curvature readout. Laboratory ice additionally sits in ambient liquid-water curvature. The pipeline books $\rho_{\text{liquid,ref}} = 1.0 \text{ g cm}^{-3}$ in $B_{\text{hom}}(\xi, \rho)$ and κ_6 second-order feedback for melt and thermodynamic witnesses (homogeneous-curvature second order, [homogeneous curvature feedback script](#)): `material_scales_bulk_h2o` exports `medium_density_fraction`; melt at $T_{\text{sl}} \approx 272.7 \text{ K}$ (vs. 273.15 K). Cell constants remain derived geometry witnesses, not NIST calibration inputs.

TABLE 4. Derived H₂O ice Ih @ 273 K, 1 atm vs. experimental benchmarks [37, 41, 38]. Column “contact lattice” includes neighbor covalent lapse overlap on the H-bond leg; “medium-closed” books ambient liquid-water curvature in $B_{\text{hom}} / \kappa_6$ (§8.2).

Quantity	Contact lattice	Medium-closed / exp.	Notes
ρ_{mass} [g/cm ³]	0.920	0.917 / 0.913	lapse overlap + Ih Bravais; PBE-D3 within 1% [43]
a [Å]	4.51	4.52 (NIST Ih)	derived Bravais from compressed r_{nn}
c [Å]	7.37	7.34 (NIST Ih)	hexagonal $c/a = 2\sqrt{6}/3$
T_{sl} [K]	272.7	273.15 (NIST)	$\kappa_6(\xi, \rho_\kappa) \times$ motif ladder
n (solid)	1.30	1.31 / 1.29	Clausius–Mossotti + η orientation
$C_{p,\text{mol}}$ [J/mol/K]	119	37.7 (ice)	contact-mode slot; structural
k_{th} [W/m/K]	1.2	2.2	phonon transport slot

Contact-lattice ρ_{mass} uses the neighbor lapse overlap slot on tetrahedral H-bond networks; remaining gaps on C_p and k_{th} flag structural slots on the same spine, not ad hoc refits.

TABLE 5. Derived H₂O allotropes @ 273 K, 1 atm (rank score from `hqiv_lab`; no DFT).

Allotrope	ρ [g/cm ³]	Score	Template
Ih (preferred)	0.920	1.80	hexagonal tetrahedral
Ic	0.95	1.52	cubic H-bond
amorphous	0.17	0.26	disordered tetrahedral

Universality across GMTKN55 condensed motifs. Water is the worked example above because ambient liquid-water curvature is the natural medium for ice-Ih melt closure. The *same* pipeline—VSEPR motif $\rightarrow$ Bravais topology $\rightarrow$ intermolecular contact r_{nn} (distinct from covalent r_{bond}) $\rightarrow$ unit cell $\rightarrow \rho_{curv} \rightarrow B_{hom} \rightarrow$ standard continuum response slots—runs on every GMTKN55 molecular spec in `hqiv_lab` without sector-specific retuning. Intermolecular contacts use only (α, γ) and steric-domain counts: H-bond path $r_{nn} = 2r_{bond} + r_{bond}\alpha(1 + \text{strongChannel})(1 + \gamma/2 \cdot \text{bend})$; apolar molecular diameter $r_{nn} = 2r_{bond}(1 + \alpha)\sqrt{1 + n_{domains} \cdot \text{strongChannel}/4}$; Bravais edges follow topology (ice Ih: $a = r_{nn}\sqrt{8/3}$, $c/a = 2\sqrt{6}/3$; fcc: $a = r_{nn}\sqrt{2}$; pyramidal fcc opens by $(1 + \gamma/4)$ for 3-fold H-bond).

Table 6 witnesses four distinct intermolecular motifs at each species’ solidification temperature: tetrahedral H-bond (H₂O @ 273 K), apolar close-pack (CH₄ @ 91 K), pyramidal H-bond (NH₃ @ 196 K), and linear chain (HF @ 190 K). Raw contact-lattice ρ_{mass} is the parameter-free geometric step in every row; ρ_{curv} clamps against the species liquid scale (1.0, 0.423, 0.682, 1.0 g cm^{−3} respectively). Refractive indices follow the same Clausius–Mossotti + binding-softness polarizability slot (§8.2), with **local-field divisors and orientation factors from optical phase participation** $\eta = \theta/\theta_0$ (TUFT bent-centre angle; local-field divisor from η , Onsager solid divisor, ice Ih orientation factor)—a drop-in replacement for per-species CM tables. Liquid disorder uses $1 - c_{Rindler} \cdot \eta$ (4/5 at $\eta = 1$); pyramidal liquids book $n_{domains}^2/4$; apolar liquids $(1 + (n_{dom} - 1)\gamma)$; linear chains $(1 + \gamma)$. Solid apolar/pyramidal uses the Onsager $n_{domains} \cdot \phi(m)/6$ slot; ice Ih orientation uses $\phi(m)/6 \cdot G_{eff}(\eta) \cdot (1 + \gamma/4)$. Apolar low- T pack uses $(T/T_{ref})^{\gamma/16}$ on r_{nn} . Liquid rows book $\rho_{liquid,ref}$ directly. Melt uses $\kappa_6(\xi, \rho_\kappa) \times$ motif ladder with $\rho_\kappa = \rho_{geom}(1 + \alpha \rho_{opt}(n))$, $\rho_{opt} = (n^2 - 1)/(n^2 + 2)$. Comparison to experiment and periodic DFT. Table 6 places HQIV witnesses beside NIST compilation targets [37, 36, 34, 35, 41] and periodic dispersion-corrected DFT at the *same* observables. Solid densities under PBE-D3(BJ) follow the ice and molecular-crystal benchmarks of [43, 5, 26]; DFT n_{solid} uses isotropic Clausius–Mossotti on static polarizabilities at the PBE-D3 geometry [39, 25]. Static lattice DFT does not supply T_{sl} without AIMD or quasi-harmonic free-energy integration [23], so the melt column compares HQIV to experiment only. HQIV uses no per-species lattice constants, dispersion parameters, or basis-set choices; the DFT columns are literature/standard-workflow references, not inputs to the spine.

Witness commands (same scripts as §8.2): `hqiv-lab CH4`, `hqiv-lab NH3`, `hqiv-lab HF` (default: species melt witness T via `hqiv_lab/species_panel.py`); [condensed phase audit script](#) exports [data/hqiv_lab-witnesses.json](#); [phase geometry density script](#) and [phase material response script](#) accept any GMTKN55 name with auto-selected preferred allotrope at the panel melt temperature. Standard continuum formulae (slot inputs). HQIV does not introduce alternate chemistries in this block: each response readout applies a *standard* textbook identity with HQIV-derived slot inputs (ρ_{curv} , B_{hom} , binding-softness polarizability, melt cohesive scale). The Clausius–Mossotti / Lorentz–Lorenz relation links n and ϵ_r to molar polarizability and density [27, 31]; phonon-limited thermal conductivity uses the kinetic transport slot [2]; molar heat capacity follows contact-mode / Debye bookkeeping [2, 3]; liquid viscosity uses the Eyring absolute-rate form [21, 3]; ice-Ih birefringence from a hexagonal c/a split is standard uniaxial optics [27, 38]. Lean mirrors (phase material response module) implement these as named slots—the continuum formulae are not HQIV-specific.

TABLE 6. Condensed-phase panel: HQIV vs. NIST vs. periodic PBE-D3 [26, 43, 5, 39]. Derived allotropes via `hqiv_lab`; no crystallographic fit in the HQIV column.

Species	Motif / al- lotrope	ρ_{solid} [g/cm ³]			n_{solid}		
		HQIV	NIST	PBE-D3	HQIV	NIST	PBE-D3
H ₂ O @ 273 K	tetrahedral / Ih	0.920	0.917	0.913	1.30	1.31	1.29
CH ₄ @ 91 K	apolar / solid_I	0.525	0.523	0.496	1.12	1.10	1.07
NH ₃ @ 196 K	pyramidal / fcc	0.816	0.817	0.795	1.29	1.32	1.27
HF @ 190 K	linear chain / Z=4	1.159	1.15	1.12	1.20	1.20	1.14
Δ / X_{NIST} (mean)		0.4% / 2.7% / 3.2%			1.2% / — / 3.3%		

Species	Motif	T_{sl}	HQIV [K]	T_{sl}	NIST [K]
H ₂ O	tetrahedral / Ih		272.7		273.15
CH ₄	apolar / solid_I		90.9		90.7
NH ₃	pyramidal / fcc		194.3		195.8
HF	linear chain / Z=4		189.1		189.6
ΔT_{sl} / T_{NIST} (mean)			0.4%		

HQIV witnesses from [phase geometry density script](#) and [phase material response script](#) at the species solidification temperature.

NIST/[41] experimental solid densities and melting points; ice $n_{\text{solid}} \approx 1.31$ [38, 37].

PBE-D3: ice Ih density [43]; CH₄, NH₃, HF lattice volumes from the molecular-crystal benchmark set [5, 26]; n_{solid} from isotropic Clausius–Mossotti on PBE-D3 static polarizabilities [39].

HQIV liquid n at the same (T, P) witness: H₂O 1.29, CH₄ 1.27, NH₃ 1.29, HF 1.15 (NIST $\approx$ 1.33, 1.29, 1.33, 1.10 respectively). Mean absolute percent errors in the summary rows compare each method column to the NIST target for that observable (PBE-D3 n row uses NIST as reference for both HQIV and DFT).

H₂O ice Ih at +0.3% after tetrahedral neighbor covalent lapse overlap $(1 - \gamma \cdot (4/8)/4)$ on the H-bond leg; CH₄ at +0.4% after apolar $(T/T_{\text{ref}})^{\gamma/16}$ contraction at melt T ; HF at +0.8% after linear monomer angle $(\pi, \text{zero bend dress})$, halogen strong H-bond leg compression $(1 - \gamma \cdot (4/8)/Z)$, and Bm21b zigzag orthorhombic open $(1 + \gamma/(6 n_{\text{inter}}))$.

Material response on the same spine. Once ρ_{curv} and the allotrope orientation are fixed, downstream readouts use binding-softness polarizability and G_{eff} contact slots (phase material response module): Clausius–Mossotti $\rightarrow n, \varepsilon_r \approx n^2$ [27]; phonon transport $\rightarrow k_{\text{th}}$ [2]; contact-mode DOF $\times B_{\text{hom}} \rightarrow C_{p,\text{mol}}$ [2, 3]; melt cohesive $\times$ shell release $\rightarrow L_{\text{fusion}}$ (structural); Eyring barrier from $E_{\text{melt}} \rightarrow$ liquid η [21, 3]; hexagonal c/a split $\rightarrow$ ice-Ih Δn [27, 38]. Witness at 273 K for derived Ih: $n \approx 1.30$, $C_{p,\text{mol}} \approx 119 \text{ J mol}^{-1} \text{ K}^{-1}$ (solid), $k_{\text{th}} \approx 1.2 \text{ W m}^{-1} \text{ K}^{-1}$, $\Delta n \approx 4 \times 10^{-4}$; liquid water $n \approx 1.33$, $C_{p,\text{mol}} \approx 74 \text{ J mol}^{-1} \text{ K}^{-1}$, $\eta \sim 10^{-4} \text{ Pa s}$ (order-of-magnitude Eyring slot). Comparison values: [37, 41, 38].

Python witnesses: `hqiv_lab/` (package; `pip install -e .`, CLI `hqiv_lab H20`), [phase geometry density script](#), [phase material response script](#), [thermodynamic phase script](#), [homogeneous curvature feedback script](#).

8.3. HQIV chem & materials lab (hqiv_lab). The `hqiv_lab` Python package is the intended home for a full “chem and materials lab”: one `MoleculeSpec` in, ranked allotropes and material properties out. The pipeline is parameter-free beyond the lattice rationals (α, γ) already fixed upstream:

- (1) `infer_monomer_geometry` — Gillespie VSEPR lone pairs and motif [24] (`tetrahedral_hbond`, `apolar_close_pack`, ...), intermolecular contact count;
- (2) `templates_for_motif` — candidate Bravais templates;
- (3) `unit_cell_for_allotrope` — (a, b, c, Z) from contact distance $\sim 2r_{\text{bond}}(1 + \alpha)$ with motif-specific opening;
- (4) `derive_allotropes` — rank @ (T, P) using the thermodynamic spine (solid bonus, melt proximity, density match to $\rho_{\text{liquid,ref}}$);
- (5) `material_response` — n, k_{th} , ... via existing script mirrors.

Lean structural mirror: phase allotrope derivation module. Extension points: custom fragment graphs (beyond GMTKN55 names), periodic contacts from [curvature contact network script](#), and glass branches from disorder scores—without reintroducing fitted intermolecular potentials.

9. OUTSIDE-CURVATURE ENVIRONMENT

9.1. Temperature slot. The outside temperature dynamics module modulates nucleon own-binding and outside caustics through outside release factor at $T(\xi) = T_{\text{Pl}}/\xi$ [17]. At lock-in the outside modulator normalises to unity; at BBN ($T \sim 0.1 \text{ MeV}$, $\xi \sim 10^{23}$) the free-branch outside-release factor is ≈ 0.785 —a $\sim 22\%$ weakening of free-branch width support relative to lock-in that feeds the coupled $\tau_n(T)$ march, neutron inventory, and ${}^7\text{Li}/H$ witnesses in the BBN and baryogenesis companions [8, 9], not Earth-surface β labs ($\sim 0.15 \text{ ppm}$ shift on τ_n at 300 K; Table 8). The faithful BBN integrator (BBN stoichiometric integrator, Lean BBN module) consumes cluster Q values from the curvature shared-well spine (curvature binding program, [data/curvature.binding.program.json](#); witness [data/bbn.integrator.json](#)), outside release, relic- ν weak-width factor, dynamic- C_2 lapse clock, and synthesis-window tail gate `bbnSynthesisDWindowTailGate` (decomposition tables via BBN paper tables script).

9.2. Gravity binding stack. Local labs sit in a cumulative weak-field potential (additive $\varepsilon = \Phi/c^2$):

$$\begin{aligned}\varepsilon_{\oplus} &= GM_{\oplus}/(R_{\oplus}c^2) \approx 7 \times 10^{-10}, \\ \varepsilon_{\odot} &= GM_{\odot}/(1 \text{ AU } c^2) \approx 10^{-8}, \\ \varepsilon_{\text{gal}} &= v_c^2/c^2 \approx 6 \times 10^{-7} \quad (v_c = 233 \text{ km/s}),\end{aligned}$$

with Galactic disk support using the same comoving baseline as the flyby readout [7]. The cumulative weak-field ledger $\varepsilon_g = \varepsilon_{\oplus} + \varepsilon_{\odot} + \varepsilon_{\text{gal}}$ is dominated by the Galactic term ($\varepsilon_{\text{gal}}/\varepsilon_g \approx 98\%$ at Earth-lab baseline): Sun and Earth contribute sub-percent fractions but are booked for cross-lab consistency. The outside gravity modulator folds this through HQIV outside G-eff support:

$$f_g = 1 + \gamma \left((1 + \varepsilon_g)^{\alpha} - 1 \right), \quad \alpha = \frac{3}{5}, \quad \gamma = \frac{2}{5}.$$

Combined lab support vs deep-space/CMB reference multiplies temperature and gravity branches. At 300 K the full stack shifts τ_n by $\sim 0.15 \text{ ppm}$ relative to zero gravity—sub-ppm and blind to bottle-beam systematics (§10); temperature dominates at $\sim 1.5 \times 10^3 \text{ ppm}$ on width readouts. The same f_g stack enters $B_{\text{curv}}(\xi) f_g$ in the free-branch opacity catalysis (§9.4) and is evaluated at hot $\xi(T)$ in the faithful BBN integrator (§11). On the *mass chart* at ξ_{lock} , Earth gravity plus

CMB-dipole proper motion contributes ~ 0.28 ppm while the inner proton anchor stays at the derived 938.272 MeV witness ([proton mass decomposition script](#)). Two labs at the same location share Sun+Galaxy in the ratio—only temperature differences remain observable on Earth.

9.3. Free nucleons: p , n , and background curvature. The light panel treats p and n as fully as the bonded isotopes, but on the *free branch* ($A \leq 1$): no nuclear well shield, unit valley geometry, and (local) relic-neutrino opacity active on the neutron (§9.4).

Neutron (n). Derived endpoint $Q_{\beta^-} = \Delta M - m_e \approx 0.782$ MeV with overlap residual ≈ 0.293 MeV. At 300 K with the full gravity stack the width ledger gives $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$ and $\tau_n \approx 880$ s ($\approx 1.00 \times$ the reference band). Outside temperature and gravity shift the clock at sub-ppm to $\sim 10^3$ ppm respectively—neither alone explains the bottle-beam split (§10).

Proton (p). The β^+ slot is kinematically **closed** at lock-in: $Q_{\beta^+} = M_p - M_n - m_e \approx -1.804$ MeV < 0 with negative effective residual; the witness reports structural stability ($\tau_p \rightarrow \infty$ after qualification). HQIV does *not* import proton-decay lifetimes as inputs.

Heat death and vanishing outside support. In the outside-temperature witness, the binding release factor at cosmological $T \rightarrow 0$ approaches ≈ 0.83 – 0.84 (not literal zero): it weakens *free-branch* binding and modulates neutron width, but does **not** open Q_{β^+} for an isolated proton because the isospin gap sign is fixed by the derived p/n masses at lock-in. A true “zero outside curvature” limit is the structural comparison point where only inside trapped curvature remains; even there the proton β^+ channel stays closed in the current $A \leq 4$ witnesses.

Neutron-star curvature (conjectural tipping scale). Regimes with lapse $N < 1$ (§6) and $\varepsilon_g \rightarrow \mathcal{O}(1)$ —orders of magnitude above the Earth-lab Galactic stack $\varepsilon_{\text{gal}} \approx 6 \times 10^{-7}$ —are where HQIV *expects* isospin tipping and $p \rightarrow n$ chemistry (neutron-star matter) to require the non-contact inverse-square / Casimir programme of §2. That density regime is **not** closed on the helium-4 panel; we record it as an invited calculation: at what ε_g (or trapped-inside ratio) the effective Q_{β^+} slot crosses zero and the proton lifetime becomes finite.

9.4. Local curvature neutrino opacity. Bound-state readouts already use the same-epoch curvature budget $B_{\text{curv}}(\xi)$ for contact networks, melt κ_6 dressing, and homogeneous second-order feedback (§8.2, homogeneous-curvature module). Ledger III inherits that stack as an *effective relic-neutrino opacity* on the weak width, not as a change to endpoint Q_{β^-} or $|\mathcal{M}|^2$ normalization.

The outer-horizon neutrino suppression $s = \text{outerHorizonNeutrinoSuppression} = 1/140$ (outer horizon neutrino suppression) dresses four weak-vertex contacts. The opacity witness is

$$(8) \quad \sigma_{\nu, \text{eff}}(\xi, \varepsilon_g) = \left(\frac{1}{s}\right)^4 B_{\text{curv}}(\xi) f_g, \quad f_g = \text{outside } G_{\text{eff}} \text{ modulator},$$

with OOM $\sigma_{\nu, \text{eff}} \approx 3.8 \times 10^8$ barn at lock-in ($B_{\text{curv}} = 1, \varepsilon_g = 0$). Lean proves opacity at lock-in = 140^4 .

Fractional catalysis on the free branch multiplies Γ_{β^-} :

$$(9) \quad \Gamma \rightarrow \Gamma \times f_w, \quad f_w = 1 + \delta_w, \quad \delta_w = \gamma^2(1-s) \left(1 - \frac{\gamma}{5}\right) \frac{4}{8} \times \left(1 + \frac{\max(B_{\text{curv}} f_g - 1, 0)}{\delta_w^{(0)}}\right),$$

with $\delta_w^{(0)}$ the lock-in catalysis at $B_{\text{curv}} f_g = 1$. A monogamy envelope $\pm \gamma/5$ on δ_w supplies error bars without method knobs. Bonded clusters ($A > 1$) **shield** relic-bath catalysis: interior valley wells already suppress weak tipping (§8); exterior caustics do not double-count the same opacity channel.

Python: [outside temperature dynamics script](#) (local ν -opacity and weak-width factor bands). The faithful BBN integrator books the same opacity on the free-neutron march ([BBN integrator script](#), [BBN condition-decay script](#); witness [BBN integrator JSON](#)).

10. PHENOMENOLOGY: LIGHT PANEL AT LOCK-IN

This section and §8.2 form the cross-domain validation layer: the *same* $(\alpha, \gamma, m_{\text{lockin}})$ ladder yields neutron lifetime, light-isotope masses, molecular shifts, and condensed-phase bulk readouts without separate sector tuning (Tables 7–4).

Witness scripts ([isotope stability script](#), [isotope PDG benchmark script](#)) export comparison tables with *reference data used only for benchmark, not for HQIV fit*. EM-tipping qualification is required for half-life claims.

TABLE 7. Light-isotope panel at lock-in (ξ lock-in, $T = 2.725$ K lab reference). Cluster masses use the curvature G_{eff} spine (§3); mass errors vs reference tables. For n , width-ledger accuracy after local ν -opacity catalysis is $\Delta Q_{\text{phase}}/Q_{\text{phase}} \lesssim 0.01\%$ (Table 9). Witness: [data/isotope_pdg_benchmark.json](#).

Isotope	(A, Z)	M_{pred}	M_{ref}	$ \Delta M /M_{\text{ref}}$	τ or width	Notes
p	(1, 1)	938.272	938.272	0.000%	—	anchor witness
n	(1, 0)	939.565	939.565	0.000%	τ ratio 1.00 ^a	free branch
D	(2, 1)	1875.651	1875.613	0.002%	stable	curvature spine
T	(3, 1)	2808.980	2808.921	0.002%	τ ratio 1.00 ^b	κ slots
He3	(3, 2)	2808.410	2808.392	0.001%	stable	
He4	(4, 2)	3727.886	3727.379	0.014%	stable	
Mean $ \Delta M /M_{\text{ref}}$				0.003%		

^a At 300 K with full gravity stack: $\tau_{\text{pred}} = 879.5$ s, $\tau_{\text{pred}}/\tau_{\text{ref}} = 1.002$; width envelope 875–884 s (Table 9).

^b Width ledger requires a distinct interior valence well (ledger locality). Without per-budget κ closure, $\tau_T/\tau_{\text{ref}} \approx 0.93$; slot-wise closure on each mass and binding budget yields 1.00 on the same equations as M_T . Collapsing mass and width into one slot (B_{cluster}/A only) gives $\approx 0.56\times$.

Post- α EC and two- α breakup. ${}^7\text{Be}$ is the first proton-rich EC/ β^+ witness on the post- α chart: endpoint $Q_{\beta^+} \approx +0.83$ MeV from local facet contact booking, valence overlap on the decaying proton slot, width well ≈ 3.16 MeV (matching $B_{\text{facet,local}}$), and post- α valley ≈ 8.5 (not $2(A-1) = 12$). The channel is structurally open; integrated τ is longer than the 53 d laboratory readout because no half-life tuning layer is applied beyond faithful slot counting (decay-chain calculator). ${}^8\text{Be}$ discharges as a broad two- α resonance on three separate ledgers (mirroring β): *binding erosion* (`multiAlphaResonanceWidth` on total B_{cluster} , BBN mass spine); *kinematic saddle* $Q_{\alpha\alpha} \approx +91$ keV from per-contact inter- α barbell booking (5 mass contacts on ΔB_{inside}); and *width* $\Gamma_{\alpha} \approx 7$ eV with shared-seam well $w_{\text{trap}} = \Delta B_{\text{inside}} \cdot 3/(2\gamma)$. Integrated $\tau \approx 6.4 \times 10^{-17}$ s (reference $\sim 6.7 \times 10^{-17}$ s). Sensitivity and error envelopes. The $\pm\gamma/5$ monogamy envelope on δ_w brackets local ν -opacity without tuning half-lives. Outside temperature dominates Earth-lab ppm budgets ($\sim 10^3$ ppm) but is sub-leading for the bottle-beam split ($\sim 10^4$ ppm); trap magnetic curvature on the weak bridge supplies the primary method ordering (§10, bottle/beam paragraph). At BBN temperatures the same $B_{\text{curv}}(\xi)$ stack weakens outside support by $\sim 22\%$ —the cosmological contrast to sub-ppm laboratory neutron clocks.

Free neutron: width ledger vs clock time. Endpoint $Q_{\beta^-} \approx 0.782$ MeV and phase-space endpoint $Q_{\text{phase}} \approx 0.782$ MeV (after m_{ν_e} and Fano/Hopf reservation). Local-curvature neutrino opacity $\sigma_{\nu,\text{eff}} \approx 3.2 \times 10^8$ barn at lab lock-in multiplies the weak width by $f_w \approx 1.073$ (envelope 1.068–1.079). Predicted $\Gamma_{\beta^-} = \ln 2/\tau_n \approx 7.88 \times 10^{-4} \text{ s}^{-1}$ ($\tau_n \approx 880$ s at 300 K). Table 9 is the preferred accuracy statement: reference experiments sit inside the catalysis envelope with $\Delta Q_{\text{phase}} \approx 0.02$ keV on the central readout—not a separate $\sim 7\%$ miss in the underlying β kernel.

TABLE 8. Outside and molecular shifts on τ readouts @ 300 K (ppm on clock time unless noted). No method knobs.

Environment / host	Observable	Shift	ppm
Isolated free n (central)	$\tau_n \approx 880$ s	ref. band	—
Full gravity stack ($\oplus + \odot + \text{Gal}$)	τ_n	+0.15	0.15
Lab temperature (300 K vs CMB)	τ_n	~ 1500	1500
Local ν -opacity catalysis	Γ_{β^-}	$\delta_w \approx 7.3\%$	7.3×10^{4a}
BBN $T \sim 0.1$ MeV	outside binding	−22%	— ^b
T ₂ per T (molecular)	τ_T	~ 30 s on 12 yr	~ 5.6
T ₂ O per T	τ_T	sub-ppm	~ 0.14

^a Fractional width catalysis; monogamy envelope $\delta_w \pm \gamma/5$ ($\approx \pm 4\%$).

^b Percent-level on binding ledger, not τ_n clock (§9).

 TABLE 9. Free-neutron accuracy in *width* units (HQIV @ 300 K, isolated atom, local ν -opacity catalysis). Envelope brackets representative bottle and beam references without method-specific knobs.

Reference	τ_{ref} (s)	τ ratio	Γ ratio	ΔQ_{phase} (keV)
PDG / CODATA	879.4	1.000	1.000	0.02
Bottle (rep.)	877.8	1.003	0.997	−0.3
Beam (rep.)	888.0	0.990	1.010	+1.5
Width envelope ($f_w = 1.073 \pm \gamma/5$)				875–884 s
$\Delta Q_{\text{phase}}/Q_{\text{phase}}$ (PDG, central)				$\lesssim 0.01\%$

No half-life parameter was tuned to match. Residuals at the central readout are within the monogamy envelope booked for weak magnetism / radiative normalization (§7.5), not a failure of the mass-budget endpoint $Q_{\beta^-} = \Delta M - m_e$.

Bottle versus beam tension (magnetic trap curvature + proved structure). This is a genuine success story for the discrete-spine approach: the $\sim 1.2\%$ world average tension is **not fitted**. HQIV predicts a central $\tau_n \approx 880$ s from Ledger III once local ν -opacity catalysis is booked; representative bottle $\tau_n^{\text{bottle}} \approx 878$ s and beam $\tau_n^{\text{beam}} \approx 888$ s (~ 10 s, $\sim 1.2 \times 10^4$ ppm) are bracketed by trap- B bridge dressing and the monogamy envelope—not by ad hoc method knobs. Lean (neutron lifetime method module) proves:

- both methods read **Ledger III** (weak β), not Ledger I (strong overlap);
- **UCN trap B** is the primary bottle dressing lever: $\rho_{\text{mag}}(B)$ on the weak Fano/Hopf bridge (trap magnetic width factor), distinct from neV Zeeman energy;
- temperature-only outside support is $\sim 10^3$ ppm—sub-leading and insufficient for the split (bottle/beam temperature insensitivity);
- the central width readout (880 s) lies between the representative references (bottle/beam reference band);
- at representative $B_{\text{bottle}} = 2.5$ T, $B_{\text{beam}} = 0$, trap dressing gives $\tau_{\text{beam}} > \tau_{\text{bottle}}$ (beam exceeds bottle from trap dressing).

Open: facility-specific B profiles, wall losses, and quantitative discharge of trap weak-bridge decoherence slot beyond the structural $f(B)$ slot.

Witness export: [isotope PDG benchmark script](#) (neutron_method_environment_ledger block).

Tritium. Same nucleon-gap endpoint Q as n . Three-ledger locality requires a distinct interior width well ($B_{\text{cluster}}/(A-1)$) blended with symmetric B_{cluster}/A , proton partners at m_p/m_n). Slot-wise local curvature on each mass and binding budget then yields $\tau_T/\tau_{\text{ref}} = 1.000$ at 300 K isolated—the *same* slot algebra that validates M_T to 0.068%. Collapsing mass and width into one ledger (B_{cluster}/A only) gives $\approx 0.56\times$, demonstrating that ledger separation is causal structure, not an optional refinement.

BBN contrast. At $T = 0.1$ MeV outside support weakens binding by $\sim 22\%$ on the free branch—the cosmological regime where percent-level effects feed Y_p , D/H , and ${}^7\text{Li}/H$ through the faithful stoichiometric integrator [8], orthogonal to Earth-lab gravity at sub-ppm. At freeze-out the same stack exports $\tau_n(T)/\tau_{n,\text{lock}} \approx 1.19$ and a weak-width catalysis factor ≈ 1.07 (`bbn_paper_tables.json`).

11. SAME EQUATIONS DOWNSTREAM: EPOCH DYNAMICS

This note **establishes** the local-curvature slot equations; baryogenesis and BBN companions **apply** them. No new binding rules are introduced in those papers—only the epoch witness and outside modulator change.

Shared slot inventory.

Scale	Local slot	Same equation
Nucleon / isotope	E_{bind} , three β ledgers, κ on budgets	§3–§8
Molecular host	ε_{mol} , bond outside contacts	§8.1
Bulk condensed phase	ρ_{curv} , $B_{\text{hom}}(\xi, \rho)$	§8.2, §8.2
Cosmological epoch	$B_{\text{curv}}(\xi(T))$, outside release, ν -opacity	§9, §9.4

Baryogenesis (companion, published). The baryogenesis lock-in note [9] (DOI [10.5281/zenodo.20711255](https://doi.org/10.5281/zenodo.20711255)) consumes lock-in η_{paper} and the same $B_{\text{curv}}(\xi)$ opacity stack exported here (§9.4, (5)). Epoch dependence enters through $T(\xi) = T_{\text{Pl}}/\xi$ and outside release at hot ξ —dynamic in cosmology, identical in slot form to the laboratory witnesses. No new binding rules or width knobs are introduced in that paper; only $\xi(T)$ and the outside modulator change. The coupled free-neutron width march at $T \sim 0.1$ MeV ($\sim 22\%$ free-branch weakening, §9) is the same Ledger III assembly documented in §7.7.

BBN abundances (companion). The BBN note [8] imports composite-trace binding for cluster Q values, dynamic- C_2 lapse suppression in the deuterium bottleneck, and stoichiometric $n+p \rightarrow D$ capture on the synthesis window $1 \text{ MeV} \rightarrow T_{\text{bn}}$, with log-weight tail `bbnSynthesisDWindowTailGate` below the ${}^3\text{He}$ mid-gate ($T_{\text{peak}} = 0.18$ MeV). This note supplies the nucleon mass spine and β -channel bookkeeping proved at Earth-lab lock-in; abundance integration lives in `hqiv_bbn_integrator.py` (witness `data/bbn_integrator.json`). The integrator books the same relic- ν opacity factor and outside-release stack as (5) on the epoch march—so ${}^7\text{Li}/H$ and the neutron inventory are not tuned independently of the Earth-lab τ_n witness. At η_{paper} the companion reports $Y_p \approx 0.244$, $D/H \approx 2.51 \times 10^{-5}$ ($z \approx -0.5$ vs. Coc), and ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-depletion; Coc semi-analytic comparison 4.65×10^{-10})—with post-BBN Li destruction documented as comparison-layer physics. Strong sector (upstream). The gluon curvature note [11] identifies strong binding with inner-Casimir trapping on the same network weights used here; continuum confinement and asymptotic-freedom language is a translation of that trapping, discharged in the upstream Lagrangian rather than imported as a separate legacy model.

12. WHAT THIS NOTE DOES NOT RE-DERIVE

The following are **not** missing physics patched in ad hoc; they are discharged upstream in [10] and companion notes, or flagged as extension milestones on the same spine. This note cites and consumes them; it does not duplicate the full Lagrangian bridge or legacy effective models.

- Continuum QCD confinement, asymptotic freedom, or a fitted strong potential as an *independent* input—strong binding is inner-Casimir trapping on the network [11, 10].
- Fermi theory, W propagators, or spectral β shapes as note-local imports—the electroweak width slot and leading $|\mathcal{M}|^2$ are upstream; forbidden transitions and radiative normalization are §7.8 extensions.
- MSbar current-quark masses or external lattice tables as inputs.
- Isotope-by-isotope half-life patches (e.g. a tritium-only factor).
- Post-BBN lithium destruction and galactic chemical evolution (pre-depletion ${}^7\text{Li}/H$ is witnessed at [8]; astrophysical depletion is not).
- Great Attractor or bulk-flow gravity layers beyond the Galactic baseline (same additive ε ledger when needed).
- Quantitative magnitudes for trap weak-bridge decoherence slot and UCN wall-loss channels (structure and directional ordering are in Lean).
- Full actinide chart integration and α -decay / fission half-life discharge (extended binding witnesses through $A = 16$ are in Table 3).
- Sub-leading inverse-square network propagation and Casimir caustics on non-contacting nucleon pairs—**conjecture only**; see §2.
- Quantitative condensed-matter tables from the raw contact lattice alone—ambient liquid-water curvature booking (§8.2) is part of the closed witness, not an optional afterthought.

13. PROGRAMME CONTEXT

Remark 13.1 (Large-scale signatures). *If lapse-selected time slicing couples coherently to the longest-wavelength background modes, one expects modified low- ℓ polarization and temperature statistics (sometimes summarized as “axis-of-evil”-type alignments). Any such prediction must eventually pass through a full Einstein–Boltzmann transport implementation; that layer is not part of this note. Separately, an algebraic theory hook relates a birefringence angle defined from $(\alpha, m_{\text{lockin}})$ to a redshift factor $(1+z) = (m_{\text{lockin}} + 1)^\alpha$ at normalization $\kappa_\beta = 1$ —a consistency display, not a claim that measured cosmic birefringence equals that value.*

Remark 13.2 (Standard Model parameter count). *Textbook electroweak fits carry on the order of twenty-six freely adjusted parameters. The HQIV presentation here reduces part of that freedom to discrete counting, a single imprint exponent forced by Lemma 3.1, and explicit structural pins (lock-in index, quark-sector anchors) whose remaining arbitrariness is acknowledged rather than hidden; the full sector projection is in [10].*

14. CONCLUSION

This note establishes the Tier-2 local-curvature foundation in the HQIV publication chain. At lock-in, nucleon masses are *constituent sums minus shared composite-trace binding* on the $\mathfrak{so}(8)$ network; the imprint ratio $\alpha = 3/5$ enters through discrete simplex counting. Weak β decay runs through three strict, non-collapsible ledgers embedded in a geometric programme of curvature balance and skew alignment (§7); the explicit width assembly (5) discharges G_F , valley exponent $2(A-1)+1$, interior valence-well blending, and free-branch $B_{\text{curv}}(\xi)$ catalysis without half-life fit knobs (§7.7). The same slot equations continue through molecular hosts and bulk condensed phases (§8.2, §8.2); baryogenesis and BBN companions apply them at cosmological $\xi(T)$ without new rules (§11).

Achieved (local-curvature spine). Through $A = 4$, radical locality on the contact network validates nucleon masses and β tipping: each ledger and isotope budget closes on its own slot, with laboratory readouts used for validation only. Machine-checked: binding positivity, ledger separation, endpoint- Q policies, width-slot wiring, Fano/Hopf bridge shape, spin-statistics quarantine, and bottle/beam method structure. Witnessed: free-neutron weak width to $\lesssim 0.01\%$ in effective phase space;

$\tau_n \approx 880$ s at 300 K; light-isotope masses at 0.003% mean; tritium $\tau_T/\tau_{\text{ref}} = 1.000$ on the same local-curvature slots as M_T (Table 7). Outside support unifies temperature and gravity on one G_{eff} primitive; bulk ρ_{curv} extends the spine to water. The faithful BBN integrator exports the same $B_{\text{curv}}(\xi)$ opacity and outside-neutron $\tau_n(T)$ stack at epoch rows—the downstream application this note prepares ([8], `data/bbn_integrator.json`).

Open (extensions on the same spine, not legacy replacements).

- **Non-contact network conjecture.** Inverse-square weights and Casimir caustics on nucleon pairs that do not share the leading composite-trace contact are *not* treated here; we invite study of whether they close sub-leading gaps (e.g. atomic vs nucleon-gap β endpoints) while preserving the three-ledger spine.
- **PDG mass gap.** Meta-harmonic nucleon readouts can sit several percent above tabulated masses until strong-sector dressing and a derived lock-in principle for m_{lockin} are closed on the upstream TUFT mass ladder [10]—the proton anchor remains a quarantined comparison witness, not an input to binding or widths.
- **Full partial widths.** Leading $|\mathcal{M}|^2$ and phase-space slots are wired from [10]; spectral shapes, forbidden transitions, and radiative normalization await the next β milestone (§7.8)—extending the upstream Lagrangian bridge, not replacing it with fitted potentials.
- m_{lockin} **without anchor.** Fixing reference $M = 4$ by the laboratory proton is consistent but not yet *derived* from a lock-in variational principle alone.
- **Nuclear chart and lithium.** Light-isotope witnesses are not a full chart; ${}^7\text{Li}/\text{H}$ is exported by `hqiv_bbn_integrator.py` on the ${}^3\text{He}+{}^4\text{He}\rightarrow{}^7\text{Be}\rightarrow{}^7\text{Li}$ ladder with trimer-width erosion, stoichiometric D budget, and synthesis-window tail gate (Lean `BBNStoichiometricIntegrator`). D/H is within $\sim 0.5\sigma$ of the observation anchor at η_{paper} (pre-CMB band).
- **Method and facility layers.** UCN wall losses, facility B maps, and quantitative trap-decoherence magnitudes beyond structural ordering.
- **Condensed-matter display layers.** Raw contact-lattice ρ_{mass} can sit $\sim 7\%$ below NIST until ambient liquid-water curvature is booked; melt and n already close on the medium-closed path (§8.2). Remaining C_p / transport gaps are structural slots on the same spine, not evidence that a separate legacy equation of state is required.

Next milestones. The β -decay programme (§7.8) extends the same upstream Lagrangian bridge to partial-width discharge and inverse β^+/EC channels. Baryogenesis and BBN companions inherit the slot inventory of §11—dynamic in epoch $\xi(T)$, identical in equation form—with Lean certificates and laboratory validation discipline unchanged.

ACKNOWLEDGMENTS

Machine-checked proofs live in the public HQIV Lean repository [20]. Reproducer scripts are documented in Appendix B. A self-contained Zenodo bundle `scripts.zip` ships every cited script plus `hqiv_lab/`; checksums are listed in the bundled `MANIFEST.sha256`. Machine-checked Lean for this note: `lake build paper_nucleon.binding` in [20].

APPENDIX A. LEAN MODULE INDEX

Anomaly cancellation (upstream):

[Hqiv/Algebra/AnomalyCancellation.lean](#).

Rapidity–Lorentz closure (upstream):

[Hqiv/Geometry/RapidityLorentzClosure.lean](#) — `rapidity_lorentz_closure_discharged`.

Spatial-rotation Lorentz closure (upstream):

[Hqiv/Geometry/SpatialRotationLorentzClosure.lean](#) — `full_lorentz_closure_discharged`.

Phase-lift skew generator:

[Hqiv/Algebra/PhaseLiftDelta.lean](#) — `phaseLiftDelta\_antisymm`.

Nucleon trace witness:

[Hqiv/Physics/BoundStates.lean](#) — triadic nucleon state on three carrier slots.

Triadic triple coupling:

[Hqiv/Physics/BoundStates.lean](#) — normalisation consumed by BBN and excited-state readouts.

BBN stoichiometric integrator:

[Hqiv/Physics/BBNStoichiometricIntegrator.lean](#) — `bbnSynthesisDWindowTailGate`, `bbnHe3SynthesisGate`, `bbnSynthesisStrongDepositionBoost`; Python mirror BBN integrator script.

Patch topology discharge:

[Hqiv/QuantumMechanics/PatchTopologicalObstruction.lean](#).

SO(8) Wilson plaquette holonomy:

[Hqiv/Physics/SO8PlaquetteHolonomy.lean](#) — `so8PlaquetteHolonomyDischarged_holds`.

Action holonomy / Wilson-kinetic glue:

[Hqiv/Physics/ActionHolonomyGlue.lean](#) — `wilsonKineticPlaquetteEquivalence_discharged`.

Discrete action Poincaré programme:

[Hqiv/Physics/DiscreteActionPoincareScaffold.lean](#) and [Hqiv/Physics/DiscreteActionStrongPoincareBridge.lean](#) — `discreteActionPoincareProgram_default`, `fullActionStrongPoincare_discharged`.

Non-abelian Haar measure scaffold (partial):

[Hqiv/Physics/NonAbelianHolonomyMeasureScaffold.lean](#).

Scale witness discipline:

`ScaleWitness`, `defaultScaleWitness` in [Hqiv/Physics/ScaleWitness.lean](#).

Nuclear inside/outside binding:

[Hqiv/Physics/NuclearCurvatureBinding.lean](#) — `nuclearClusterBindingCurvature`, `nuclearValleyContactCount`.

HQIV nuclei: Fresnel caustics and ladder:

[Hqiv/Physics/HQIVNuclei.lean](#) — `fresnelCaustic`, `valleyCount`, `deuteron`, `helium4`, `helium4\_valleyCount`.

Nuclear contact closure / zero deficit:

[Hqiv/Physics/NuclearContactClosure.lean](#) — `contactInformationDeficit`, `isFullyContactClosed`, `fullyInteractingAlphaWitness`, `first\_zero\_deficit\_mass\_number\_eq\_referenceM`.

Casimir caustic hierarchy (Python):

[scripts/hqiv_nuclear_caustic_binding.py](#) — `sphere_pair_overlap`, `barbell_torus`, `torus_deepen_step*`, `tetrahedral_closure`.

Bound states / network spine:

[Hqiv/Physics/BoundStates.lean](#) — `E\_bind\_from\_network`, `E\_bind\_from\_composite\_trace`.

Nucleon shared binding:

`nucleonSharedBinding\_MeV`, `nucleonSharedBinding\_from\_composite\_trace`.

Derived nucleon mass module:

[Hqiv/Physics/DerivedNucleonMass.lean](#).

Derived nucleon mass pack:

`derivedProtonMass\_pos`, `derivedNeutronMass\_pos`, `constituent\_isospin\_splitting`.

Isospin gap unit:

`nucleonIsospinGap\_eq\_one` in [Hqiv/Physics/QuarkMetaResonance.lean](#).

Proton anchor condition:

`ProtonAnchorCondition`, `proton\_anchor\_condition\_discharge`.

Neutron stability scaffold:

[Hqiv/Physics/NeutronBindingStabilityScaffold.lean](#) — `freeNeutronStrongDecayWidth`, `freeNeutronWeakDecayWidth`, `free\_neutron\_strong\_half\_life\_from\_spin\_statistics`.

Spin-statistics module:

[Hqiv/Physics/SpinStatistics.lean](#) — exchange phase -1 for half-integer modes.

Spin-statistics half-life:

`spin\_statistics\_determines\_half\_life`, `resonance\_half\_life` in [Hqiv/Physics/HQIVNuclei.lean](#).

Dynamic β isotope module:

[Hqiv/Physics/DynamicBetaIsotope.lean](#) — `betaMinusMassGapAtXi`, `betaMinusCurvatureOverlap`, `betaMi`

nusResidualAtXi, betaMinusEndpointQAtXi, betaWeakMatrixElementSquared, betaGeometryWidthFactor, betaWeakWidthFromResidual (see module for full list).

Endpoint- Q identity:

betaMinusEndpointQAtXi\eq_derivedDeltaM_minus_m_e.

Curvature mass imprint:

curvatureMassImprint, imprintedMassBudget, imprintedMassBudget\eq_ref in [Hqiv/Physics/DynamicBetaIsotope.lean](#).

Nucleon-gap endpoint policy:

betaMinusEndpointQNucleonGap, betaMinusEndpointQNucleonGap\eq_betaMinusEndpointQAtXi, betaMinusIsotopeMassGapAtXi\eq_betaMinusMassGapAtXi.

Uniform imprint preserves ΔM :

betaMinusEndpointQUniformImprint, betaMinusEndpointQUniformImprint\eq_kappa_gap_minus_m_e.

Per-isotope imprint $\rightarrow$ mass-table Q :

betaMinusEndpointQPerIsotopeImprint, betaMinusEndpointQPerIsotopeImprint\eq_reference_budgets in [Hqiv/Physics/DynamicBetaIsotope.lean](#).

Geometric-mean width well:

betaWidthWellGeometricBlend, widthWellCurvatureImprint.

Valley count bound:

betaValleyCountBound, betaCausticLayerCount.

Weak width slot:

betaWeakWidthSlot, betaWeakWidthSlot_betaMinus.

Fermi coupling from β :

G\F_from_beta in [Hqiv/Physics/Forces.lean](#).

Beta decay rate scaffold:

beta_decay_rate in [Hqiv/Physics/NuclearAndAtomicSpectra.lean](#).

Weak beta channel open:

weakBetaChannelOpen in [Hqiv/Physics/DynamicIsotopeStability.lean](#).

Beta width ledger qualified:

betaWidthLedgerQualified, EMTippingQualified in [Hqiv/Physics/DynamicIsotopeStability.lean](#).

EM tipping gate:

Legacy export name EMTippingQualified = betaWidthLedgerQualified at m_e_PDG; stability uses structurallyShieldedIsotope.

Neutron lifetime method (bottle/beam):

[Hqiv/Physics/NeutronLifetimeMethod.lean](#) — trapMagneticCurvatureFraction, trapWeakWidthFactorFromMagnetic, beam_tau_gt_bottle_from_magnetic_trap_dressing, temperature_ppm_insufficient_for_bottle_beam_split, TrapWeakBridgeDecoherenceSlot, central_brackets_bottle_beam_refs.

Dynamic isotope stability:

[Hqiv/Physics/DynamicIsotopeStability.lean](#) — structurallyShieldedIsotope, dynamicallyStableIsotope, betaHalfLifeFromWidth\eq.

Half-life from width:

betaHalfLifeFromWidth\eq.

Weak Fano/Hopf bridge:

[Hqiv/Physics/WeakFanoHopfBridge.lean](#) — weakBridgeShape, weakBridgeEnergyMeV, defaultBetaWeakBridge.

Outside temperature dynamics:

[Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#) — T_MeV_from_xi, outsideCurvatureReleaseFactor, nucleonOwnBindingAtXi, betaMinusOverlapAtXi, localCurvatureNeutrinoOpacityBarn, localCurvatureWeakWidthFactor (see module for full list).

Local ν opacity at lock-in:

localCurvatureNeutrinoOpacityBarn_lockin_zeroGravity (140⁴ barn OOM).

Outer horizon neutrino suppression:

outerHorizonNeutrinoSuppression, outerHorizonNeutrinoSuppression\eq_inv_140 in [Hqiv/Physics/DerivedGaugeAndLeptonSector.lean](#).

Outside release factor:

`outsideCurvatureReleaseFactor\_pos`.

Outside G_{eff} support:

[Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#) — `OutsideGravityWitness`, `outsideEnvironmentModulator` (see module for full list).

Bond-state network module:

[Hqiv/QuantumChemistry/BondStateNetwork.lean](#), [Hqiv/QuantumChemistry/CurvatureBondContact.lean](#).

Homogeneous-curvature second order:

[Hqiv/Physics/HomogeneousCurvatureSecondOrder.lean](#) — `homogeneousCurvatureBudgetAtXi`, `effectiveCurvatureBudgetAtXi` (see module for full list).

Phase allotrope derivation (hqiv.lab):

[Hqiv/QuantumChemistry/PhaseAllotropeDerivation.lean](#) — `IntermolecularMotif`, `AllotropeCandidate`, `DerivedUnitCell`, `motif\_h2o\_has\_ice\_ih\_label`.

Phase geometry $\rightarrow$ density $\rightarrow \rho_{\text{curv}}$:

[Hqiv/QuantumChemistry/PhaseGeometryDensity.lean](#) — `PhaseUnitCell`, `curvatureDensityFraction`, `phaseUnitCellH2OIceIh` (see module for full list).

Phase material response (n , ε_r , k_{th}):

[Hqiv/QuantumChemistry/PhaseMaterialResponse.lean](#) — `clausiusMossottiRatio`, `refractiveIndexFromCM`, `localFieldDivisorFromEta`, `solidOnsagerLocalFieldDivisor`, `phaseOrientationCmFactorTetrahedralIh`, `gEffFromPhaseParticipation`, `molarHeatCapacitySlot`, `latentHeatFusionSlot`, `birefringenceDeltaNSlot`, `dynamicViscositySlot` (see module for full list).

Outside molecular witness:

`OutsideMolecularWitness` in [Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#).

Dynamic nucleon p/n pair:

[Hqiv/Physics/DynamicNucleonPN.lean](#).

Caustic binding stack:

[Hqiv/Physics/NuclearCausticBinding.lean](#), [Hqiv/Physics/NuclearCurvatureBinding.lean](#).

Nuclear spectra module:

[Hqiv/Physics/NuclearAndAtomicSpectra.lean](#).

Isotope valley scaffold:

[Hqiv/Physics/HQIVNuclei.lean](#) — `valleyCount`, `valleys\_are\_additive`.

APPENDIX B. WITNESS SCRIPTS AND JSON EXPORTS

Isotope stability half-life script:

[scripts/hqiv_isotope_stability_halfife.py](#) — stability readout, half-life, outside support; [data/isotope_stability_halfife.json](#).

Dynamic beta isotope script:

[scripts/hqiv_dynamic_beta_isotope.py](#) — three-ledger β channel chart; [data/dynamic_beta_isotope.chart.json](#).

Isotope PDG benchmark script:

[scripts/hqiv_isotope_pdg_benchmark.py](#) — light-panel benchmark table; [data/isotope_pdg_benchmark.json](#).

Outside temperature dynamics script:

[scripts/hqiv_nuclear_outside_temperature_dynamics.py](#) — outside $T(\xi)$ and gravity binding stack.

Curvature binding program script:

[scripts/hqiv_curvature_binding_program.py](#) — preferred G_{eff} + shared-well binding witness; [data/curvature_binding_program.json](#).

Binding energy program script:

[scripts/hqiv_binding_energy_program.py](#) — unified constructive + post- α panel; [data/binding_energy_program.json](#).

Nuclear caustic binding script:

[scripts/hqiv_nuclear_caustic_binding.py](#) — diagnostic outside-caustic layer stack (not PDG-comparable totals).

Inside/outside binding script:

[scripts/hqiv_nuclear_inside_outside_binding.py](#).

Dynamic nucleon p/n script:

[scripts/hqiv_dynamic_nucleon_pn.py](#).

Bond-state network script:

[scripts/hqiv_bond_state_network.py](#) — molecular bond-state network; T₂, T₂O hosts.

Materials lab package (hqiv_lab):

derived allotropes, hqiv-lab CLI; install via `pip install -e ..`

Phase geometry density script:

[scripts/hqiv_phase_geometry_density.py](#) — ρ_{curv} melt readout via hqiv_lab.

Phase material response script:

[scripts/hqiv_phase_material_response.py](#) — n , ε_r , k_{th} , C_p , L_f , η , Δn .

Condensed phase audit script:

[scripts/hqiv_condensed_phase_audit.py](#) — GMTKN55 panel @ species melt T ; [data/hqiv_lab_witnesses.json](#).

Thermodynamic phase script:

[scripts/hqiv_thermodynamic_phase_from_tp.py](#).

Homogeneous curvature feedback script:

[scripts/hqiv_homogeneous_curvature_feedback.py](#).

Curvature contact network script:

[scripts/hqiv_curvature_contact_network.py](#).

BBN integrator script:

[scripts/hqiv_bbn_integrator.py](#) — [data/bbn_integrator.json](#).

BBN condition-decay script:

[scripts/hqiv_bbn_condition_decay.py](#).

BBN paper tables script:

[scripts/hqiv_bbn_paper_tables.py](#) — [data/bbn_paper_tables.json](#).

Integrator Lean audit script:

[scripts/hqiv_integrator_lean_audit.py](#) — [data/integrator_lean_audit.json](#).

Decay-chain calculator:

[scripts/hqiv_decay_chain.py](#), [scripts/hqiv_decay_calculator.py](#) — nuclear $\beta^\pm$, EC/ β^+ (post- α local contact), and two- α breakup on the canonical binding spine.

Post- α binding program:

[scripts/hqiv_post_alpha_binding_program.py](#), [scripts/hqiv_post_alpha_sphere_touching.py](#).

Phase material response tests:

[scripts/test_hqiv_phase_material_response.py](#).

Run from repository root (see bundled [scripts/README.md](#) in [scripts.zip](#)):

```
pip install -e .
```

```
hqiv-lab H2O
```

```
hqiv-lab CH4 --json
```

```
python3 scripts/hqiv_isotope_pdg_benchmark.py --qualify-em-tipping
```

```
python3 scripts/hqiv_isotope_stability_half-life.py --A 1 --Z 0 \
```

```
--qualify-em-tipping --lab-temperature-K 300 --gravity-stack full
```

```
PYTHONPATH=scripts python3 scripts/hqiv_bbn_integrator.py
```

```
PYTHONPATH=scripts python3 scripts/hqiv_bbn_paper_tables.py
```

```
PYTHONPATH=scripts python3 scripts/hqiv_decay_calculator.py nuclear --seed Z4A7 --max-depth 3
```

```
PYTHONPATH=scripts python3 scripts/hqiv_decay_calculator.py nuclear --seed Z4A8 --max-depth 2
```

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